PREPRINT

Exploring Equivalence Testing with the Updated TOSTER R Package

Aaron R. Caldwell[a]

[a]Department of Biostatistics, Fay W. Boozman College of Public Health, University of Arkansas for Medical Sciences Northwest, Springdale, AR,
https://orcid.org/0000-0002-4541-6283

ARTICLE HISTORY
Compiled October 13, 2026

ABSTRACT
Despite decades of methodological guidance, researchers continue to misinterpret non-significant results as evidence for the absence of an effect. Equivalence testing via two one-sided tests (TOST) provides a principled alternative, but adoption remains limited by software accessibility and lack of training. The TOSTER R package, originally developed by Lakens (2017), made TOST accessible to experimental psychologists. This paper describes a major update to the package that expands its scope with (1) a unified t-test interface supporting both equivalence and minimal effect testing, (2) a simplified hypothesis testing interface returning standard R objects, (3) tools for computing and testing standardized effect sizes and correlations, (4) robust and resampling-based alternatives including bootstrap and permutation tests, (5) equivalence testing for ANOVAs, and (6) power analysis and sample size planning. Each capability is demonstrated with worked examples. TOSTER is also available as a jamovi module for researchers who prefer a graphical interface.

KEYWORDS
statistics, bootstrap, minimal effects test, NHST, TOST, R package, robust statistics

1. Introduction

Researchers routinely interpret non-significant p-values from a nil-hypothesis (i.e., null hypothesis that the mean difference equals zero) test as evidence that no effect exists — a logical error that has been recognized for decades (Altman and Bland 1995). A non-significant result means only that the data are compatible with the null hypothesis; it does not mean the null hypothesis is true.

Equivalence testing provides a principled framework for evaluating whether an effect is small enough to be considered negligible. The two one-sided tests (TOST) procedure is the most widely used approach (Schuirmann 1987). Researchers specify an upper (Δ_U) and lower (Δ_L) equivalence bound based on the smallest effect size of interest (SESOI). Two one-sided tests then evaluate whether the observed effect falls within those bounds. If both tests reject at a pre-specified alpha level, the effect can be considered practically equivalent to zero (Lakens 2017).

CONTACT Aaron R. Caldwell. Email: caldwellaaron@uams.edu

The TOSTER R package was originally developed by Lakens (2017) to introduce experimental psychologists to the concept of equivalence testing and provide an accessible implementation in R. The package provided dedicated functions for independent, paired, and one-sample t-tests, as well as correlations and meta-analytic equivalence tests. It has been widely adopted, with over 1000 citations to date.

However, the original package had a number of limitations. It offered a function-per-design interface (separate functions for each t-test type), lacked robust alternatives, provided no support for ANOVA equivalence testing, and had no tools for comparing effect sizes across studies. Also, the package did not integrate with standard R workflows and usually just printed results to the console rather than returning objects that could be manipulated or plotted.

This paper provides a tutorial-style introduction to the updated TOSTER package, organized around its major capabilities: (1) a unified t-test interface (`t_TOST`) that pairs a standard significance test with an equivalence test and supports minimal effect testing, (2) a simplified hypothesis testing interface (`simple_htest`) that returns standard `htest` objects, (3) tools for computing and testing standardized effect sizes — including standardized mean differences (SMDs), rank-based effect sizes, and correlations, (4) robust and resampling-based alternatives including the Brunner-Munzel test, bootstrap and permutation t-tests, and log-transformed methods, (5) equivalence testing for ANOVAs, and (6) power analysis and sample size planning. Each section motivates when and why a researcher would use TOSTER's functions before demonstrating how TOSTER actually works.

TOSTER occupies a specific niche in the R ecosystem in that it is a frequentist toolkit for equivalence testing of standard designs. Researchers interested in Bayesian approaches to evidence for the null have excellent options (e.g., "bayestestR" by Makowski, Ben-Shachar, and Lüdecke (2019)). For equivalence-style contrasts in more complex models (e.g., mixed models) the "emmeans" (Lenth and Piaskowski 2026) and "marginaleffects" (Arel-Bundock, Greifer, and Heiss 2024) packages can accommodate non-zero null hypotheses. TOSTER fills the space of accessible, self-contained equivalence testing for common, albeit simple, research designs. For a broader introduction to the methodology, see Lakens (2017), Lakens, Scheel, and Isager (2018), Lakens et al. (2020), Mazzolari et al. (2022), and Tomek, Caldwell, and Eisner (2026). TOSTER is also available as a jamovi module for researchers who prefer a graphical interface although it has limited capabilities compared to the R package.

2. Setting Equivalence Bounds

There are two decisions that I think warrant consideration when using equivalence testing. The first is to formally define the quantity specified by the hypothesis, that is, to choose an estimand. The second is to determine how much of that quantity is too small to matter, that is, to choose the equivalence bounds. These decisions are interdependent. A bound is a magnitude on the scale of some particular quantity (i.e., estimand), and the same numerical bound can therefore express quite different hypotheses depending on what is being estimated.

2.1. *Choosing an Estimand*

The estimand is the quantity a study is attempting to estimate (Lundberg, Johnson, and Stewart 2021). In an ideal scenario, the hypothesis should determine the estimand,

and estimand then narrows the set of appropriate tests. Not considering the estimand is a reliable way to end up testing the wrong hypothesis.

This matters more for equivalence testing than for nil-hypothesis testing. When two distributions are identical, the location-type quantities in Table 1 all take their null values at once (the mean difference and the pseudo-median are both zero, and the probability of superiority is exactly 0.5) so a nil test is comparatively forgiving of an unexamined estimand. An equivalence bound makes the decision more consequential. It is a specific magnitude on a specific scale, and there is generally no way to translate a bound on one scale into a bound on another (at least without making very tenuous assumptions). Bounds of $\pm$ 0.5 placed on a mean difference, on a median/pseudo-median, and on the SMD describe three different hypotheses, only one of which is likely to be the intended one.

Table 1 lists the estimands TOSTER can place bounds on and the functions that target each. The trimmed means option, available through the `tr` argument of the resampling functions (`tr = 0` by default), coincides with the mean only when the distribution is symmetric. Therefore, some thought should be given to the fact that the the estimand changes when trimmed means are utilized, and it is not just making the same estimate more robust. Additionally, all SMDs are a rescaling of the mean difference (much like a z-score) by a sample-dependent quantity (caution about standardizing using SMDs is given later).

Table 1.: Estimands supported by TOSTER, the scale on which equivalence bounds are specified for each, and the primary functions targeting them.

Estimand	Scale of the bound	Primary functions
Mean or mean difference	Original units	`t_TOST`, `tsum_TOST`, `simple_htest`
Mean, with empirically calibrated reference distributions	Original units	`boot_t_TOST`, `boot_t_test`, `perm_t_test`
Trimmed mean difference	Original units	`boot_t_TOST`, `perm_t_test` (via `tr`)
Ratio of geometric means	Multiplicative ratio	`log_TOST`, `boot_log_TOST`
Pseudo-median (Hodges-Lehmann)	Original units	`wilcox_TOST`
Stochastic superiority	Probability (0 to 1)	`brunner_munzel`
Standardized mean difference	Standard deviation units	`smd_calc`, `boot_smd_calc`
Correlation	Correlation (-1 to 1)	`z_cor_test`, `boot_cor_test`
Proportion of variance explained	Proportion (0 to 1)	`equ_anova`, `equ_ftest`

In applied settings the estimand is not fully pinned down by the choice of summary statistic either. Decisions about how intercurrent events (post-randomization events such as dropout) and missing data are handled determine which population summary is being estimated at all. Two analyses that both report "the mean difference" may be estimating materially different things if they resolve these questions differently.

It is prudent to settle on the estimand first and then justify a bound on its scale, rather than the dubious process selecting a test from decision tree and adopting whatever quantity it happens to return. This estimand-first style approach also makes the robust methods described later in this paper easier to navigate. The robust methods discussed here are often tests of different quantities, and only some of them remain hypothesis tests of means.

2.2. *Justifying the Bounds*

Before applying any equivalence test, the researcher must specify the equivalence bounds which defines the range of effect sizes considered too small to be meaningful. There is no single correct method for choosing these bounds; the appropriate approach depends entirely on the research question and domain of research.

If a researcher has an equivalence hypothesis, they should be able to articulate what magnitude of effect would be too small to matter in context. The SESOI must flow from the hypothesis formulated by theory and context not from a formula. However, there are several general strategies that have been evaluated (Cook et al. 2018). Anchor-based methods tie the bound to an externally meaningful criterion, such as a clinically meaningful difference or a "just-noticeable" difference (Anvari and Lakens 2021). Distribution-based methods define the bound as a fraction of the measurement error (Jacobson and Truax 1991), though this approach is generally not recommended (Cook et al. 2018). Cost-benefit, also referred to as health economic methods (Cook et al. 2018), reasoning sets the bound at the point where there is a net benefit relative to the cost of an intervention. Researchers can draw on existing literature or normative data to identify effect magnitudes that prior work suggests are negligible. When theory is sufficiently precise, however, the bound may require no external benchmark at all, and the hypothesis itself implies what magnitude of effect would be too small to matter.

Using "default" bounds on the SMD (also known as Cohen's d) based on Cohen's benchmarks (e.g., SMD = 0.2, 0.5, 0.8 to imply small, medium, or large effects, respectively) without domain-specific justification is strongly discouraged. Cohen himself stated that these conventions were "recommended for use only when no better basis for estimating the index is available" (p. 25, as cited in Panzarella, Beribisky, and Cribbie (2021)) and reportedly came to regret having proposed them at all (Funder and Ozer 2019). These benchmarks should never be used as universal thresholds. When applied uncritically, these benchmarks can produce equivalence bounds that are too wide or too narrow, and likely entirely disconnected from the substantive research question.

If a researcher cannot specify meaningful bounds, this itself is informative as it suggests the equivalence hypothesis is not well enough developed to test such a hypothesis. In such cases, it may be more productive to focus on effect estimation and description rather than leaping to hypothesis tests (Scheel et al. 2020). Throughout the examples in this paper, equivalence bounds, and the data sets themselves, are chosen for illustrative purposes; readers should not treat them as direct recommendations for their own research.

3. TOST with t-tests

The t-test is the standard tool for comparing means. We can estimate a mean (one-sample), the mean of the paired differences (paired samples), or the difference in means

between two groups, and then divide that estimate by its standard error. The resulting t-statistic is compared against a t-distribution to obtain p-values and confidence intervals. Written generally, the statistic takes the form $t = (\hat{\mu} - \mu_0)/SE_{\hat{\mu}}$, where $\hat{\mu}$ is the estimate and μ_0 is the value being tested against.

What matters for equivalence testing is that μ_0 is a choice. A traditional nil hypothesis test fixes $\mu_0 = 0$ and asks whether the mean, or mean difference, is exactly zero, with a "significant" difference declared for any departure from this point null hypothesis. TOST only changes what is substituted for μ_0. Two one-sided tests are performed — one against the lower bound ($\mu_0 = \Delta_L$) and one against the upper bound ($\mu_0 = \Delta_U$) — and equivalence is concluded only if both reject. The null hypothesis is no longer a single point but a composite one (the effect lies at or beyond either bound), and the equivalence region between the bounds becomes the alternative. Because each test is one-sided at α, the interval estimate corresponding to the procedure is a $1 - 2\alpha$ (typically 90%) confidence interval, which is what TOSTER reports and can plot.

Throughout, the estimand for a t-test is the mean, or mean difference, on the original scale of measurement (or, more precisely, on whatever scale the t-test is calculated, such as log-transformed data). As discussed above and described later in this paper, many robust "alternatives" to the t-test estimate different quantities entirely, and are not interchangeable substitutes when the hypothesis concerns means.

The `t_TOST` function in TOSTER provides a unified interface for equivalence and minimal effect testing with t-tests. It runs a standard nil significance test alongside the equivalence (or minimal effect) test, because in practice researchers may want both: "Is there a significant effect?" and "Can I rule out a meaningful effect?" The function handles independent, paired, and one-sample designs through the `paired` argument and accepts data via either a formula interface or raw vectors, making it a direct analog to base R's `t.test`. Additionally, the output includes an SMD calculation alongside the unstandarized mean difference with appropriate, albeit approximate, confidence intervals. For researchers who only want to test a single hypothesis at a time, the `simple_htest` function (described in the next section) provides a lighter alternative.

Minimal effect testing (MET) is also supported. MET is useful when the hypothesis is that an effect *exceeds* a meaningful threshold. In the case of MET, the null hypothesis is equivalence, and rejection implies a practically meaningful effect (see Figure 1) (Mazzolari et al. 2022).

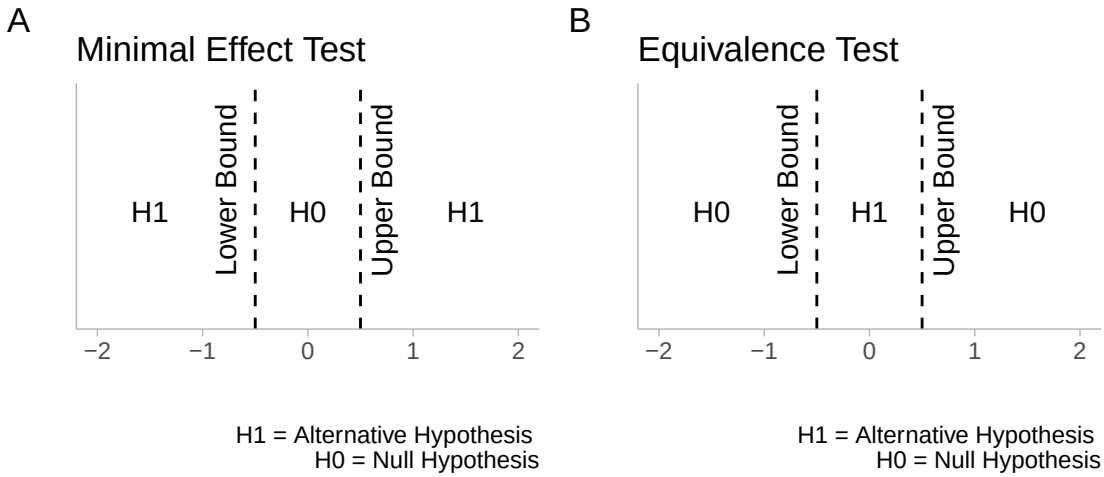


Figure 1. Types of hypotheses tests supported by two one-sided tests (TOST) procedures.

3.1. *Independent Groups Example*

Throughout this manuscript, examples use two datasets. The `bugs` data from the `jmv` R package contains participants' ratings (on a 0–10 scale) of their desire to kill bugs that vary in how disgusting and frightening they are, and is used for the independent-groups and one-sample examples. The `sleep` data, which records the increase in hours of sleep for the same 10 patients under two drugs, is used for the paired examples in later sections. Because one participant in `bugs` has no recorded gender, I first restrict the data to participants who reported either female or male gender.

```
data('sleep')
library(jmv)
data('bugs')
# keep participants with a recorded gender
bugs_g = droplevels(subset(bugs, Gender %in% c("Female", "Male")))
```

For the first example, we can compare female and male participants' ratings for the low disgust, high fear (`LDHF`) bug with equivalence bounds (set by the `eqb` argument) of ± 1 point on the rating scale. The `smd_ci` argument can also be set to use one of four possible calculations for the SMD confidence interval. In the example below, I have opted to go with the normal approximation for computational ease; such approximations tend provide adequate coverage (Viechtbauer 2007). The formula interface works almost identically to `t.test`:

```
res1 = t_TOST(
  # difference in "Gender" of outcome "LDHF"
  formula = LDHF ~ Gender,
  # data
  data = bugs_g,
  # equivalence bound
  eqb = 1,
```

```
  # sets type of SMD confidence interval
  smd_ci = "z"
)
```

The `print` method provides a verbose summary including the two-tailed test, both one-sided equivalence tests, and the SMD (Hedges' g by default):

```
print(res1)
```

```
##
## Welch Two Sample t-test
##
## The equivalence test was non-significant, t(58.15) = -0.87, p = 0.2
## The null hypothesis test was non-significant, t(58.15) = 0.921, p = 0.36
## NHST: don't reject null significance hypothesis that the effect is equal to zero
## TOST: don't reject null equivalence hypothesis
##
## TOST Results
##                  t    df p.value
## t-test      0.9213 58.15   0.361
## TOST Lower  2.7084 58.15   0.004
## TOST Upper -0.8658 58.15   0.195
##
## Effect Sizes
##                Estimate     SE              C.I. Conf. Level
## Raw              0.5155 0.5596 [-0.4198, 1.4508]         0.9
## Hedges's g(av)   0.2037 0.2270 [-0.1697, 0.5771]         0.9
## Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
## Equivalence Bounds: Raw [-1, 1]; Hedges's g(av) [-0.3989, 0.3989]
```

A greater description of the results can be accomplished with `describe` method.

```
cat(describe(res1))
```

Using the Welch Two Sample t-test, a null hypothesis significance test (NHST), and a equivalence test, via two one-sided tests (TOST), were performed with an alpha-level of 0.05. These tested the null hypotheses that true mean difference is equal to 0 (NHST), and true mean difference is more extreme than -1 and 1 (TOST). Both the equivalence test (p = 0.195), and the NHST (p = 0.361) were not significant (mean difference = 0.516 90% C.I.[-0.42, 1.451]; Hedges's g(av) = 0.204 90% C.I.[-0.17, 0.577]). Therefore, the results are inconclusive: neither null hypothesis can be rejected.

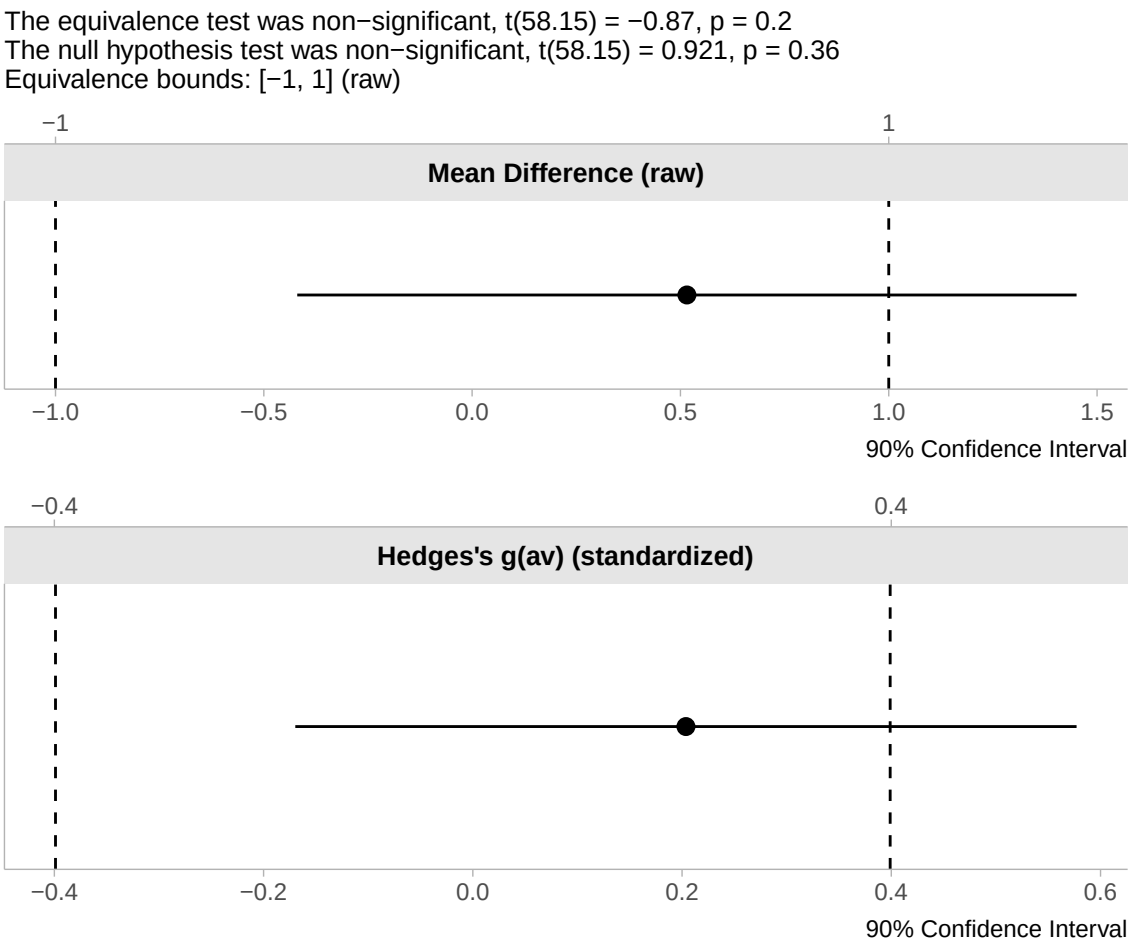


Figure 2. Results of the independent-samples equivalence test visualized with the default plot method. The top panel displays the raw mean difference with its 90% confidence interval, and the bottom panel displays the standardized effect size (Hedges's g_{av}) with its 90% confidence interval. In both panels, the dashed vertical lines indicate the equivalence bounds (raw: -1 to 1; standardized: approximately -0.4 to 0.4), and the point estimate is marked by a filled circle. A 90% confidence interval is used because it corresponds to the two one-sided tests (TOST) procedure at an alpha level of 0.05.

The default visualization provided by the generic `plot` method for `t_TOST` provides a simple visualization of both effect sizes and their respective confidence intervals alongside the equivalence bounds. Test information is also printed at the top of the plot.

```
plot(res1)
```

Also when the generic `plot` method has the `type` argument set to "cd" this produces a consonance density plot, inspired by the "concurve" R package (Rafi and Greenland 2020), that shows the estimate, confidence intervals, and equivalence bounds simultaneously:

```
plot(res1, type = "cd")
```

Other plot types are available: `type = "c"` produces a consonance plot showing

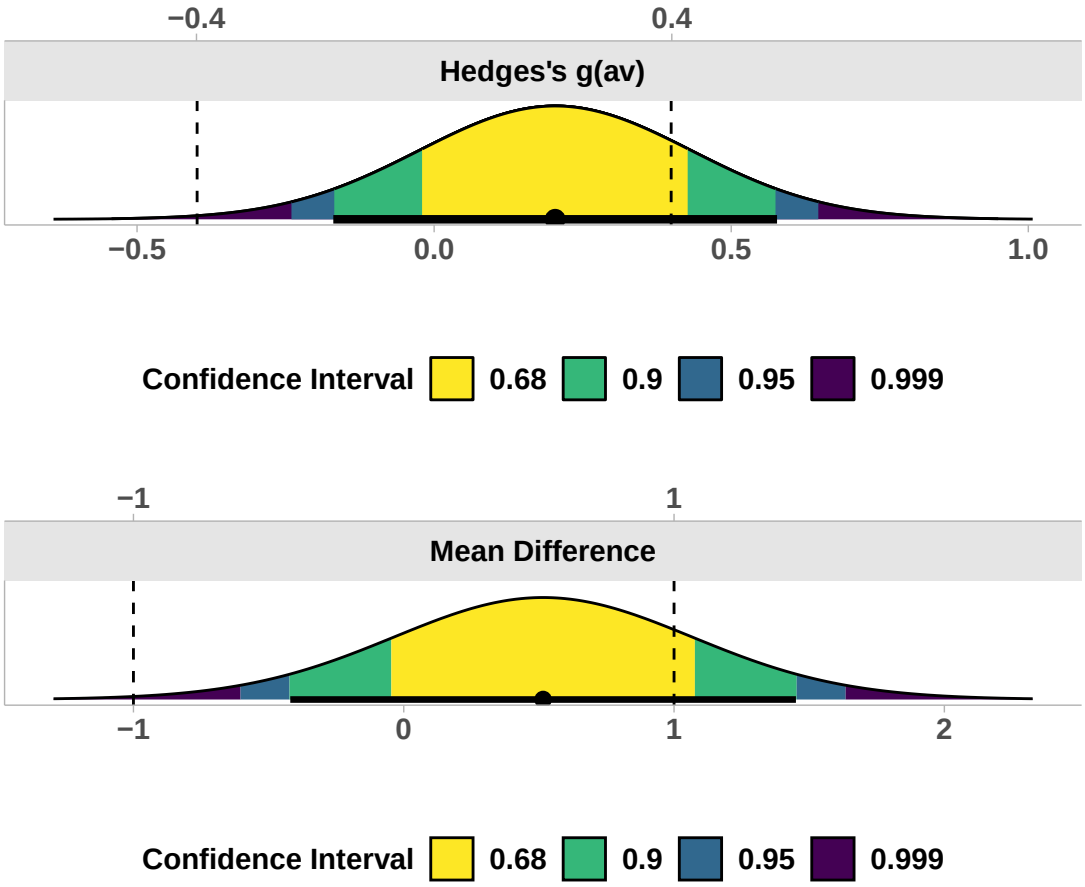


Figure 3. Consonance density plot for the independent-samples equivalence test. The top panel displays the standardized effect size (Hedges's g_{av}) and the bottom panel displays the raw mean difference. In each panel, the consonance density curve represents the distribution of confidence intervals across all levels, with color-coded regions corresponding to the 68%, 90%, 95%, and 99.9% confidence intervals. The point estimate is marked by a filled circle along the x-axis. The dashed vertical lines indicate the equivalence bounds (standardized: approximately -0.4 to 0.4; raw: -1 to 1). This visualization conveys the full range of effect sizes compatible with the data at varying confidence levels, rather than relying on a single confidence interval. Wider, less certain intervals (e.g., 99.9%) extend further from the point estimate, while narrower intervals (e.g., 68%) cluster near it.

multiple confidence interval levels, and shading can be customized with the `ci_shades` argument.

3.2. *One-Sample t-test with Asymmetric Bounds*

When the equivalence bounds are not symmetric around the null, two values can be supplied to `eqb`. This is common when the research question implies asymmetric consequences. Similarly, we may want to test the mean itself rather than a mean difference. This harkens back to the original use of the t-test by Gosset at Guinness Brewery, where the concern was whether a single sample mean fell within acceptable limits. Here we test whether the mean LDHF rating falls between 5.5 and 8.5, while the nil hypothesis test is performed against a hypothesized value of 7.5 (set with the `mu` argument). Note that the equivalence bounds are specified on the original scale of the data (i.e., they are values of the mean, not distances from `mu`), and the raw estimate and its confidence interval are reported on that same scale. The SMD, in contrast, is calculated relative to `mu` (i.e., $(\bar{x} - \mu)/s$), so its bounds are expressed as standardized distances from `mu`; the printed output notes this whenever `mu` is not zero:

```
res4 = t_TOST(
  # single sample, vector from which to estimate mean
  x = bugs$LDHF,
  # set nil significance test to 7.5
  mu = 7.5,
  # set TOST bounds to 5.5 and 8.5
  eqb = c(5.5, 8.5)
)
res4
```

```
##
## One Sample t-test
##
## The equivalence test was significant, t(90) = -4.2, p < 0.01
## The null hypothesis test was non-significant, t(90) = -0.458, p = 0.65
## NHST: don't reject null significance hypothesis that the effect is equal to 7.5
## TOST: reject null equivalence hypothesis
##
## TOST Results
##                  t df p.value
## t-test     -0.4577 90   0.648
## TOST Lower  7.1156 90 < 0.001
## TOST Upper -4.2444 90 < 0.001
##
## Effect Sizes
##            Estimate     SE              C.I. Conf. Level
## Raw         7.37912 0.2641   [6.9402, 7.818]         0.9
## Hedges's g -0.04758 0.1049 [-0.2185, 0.1236]         0.9
## Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
## Equivalence Bounds: Raw [5.5, 8.5]; Hedges's g [-0.7939, 0.397]
## Note: raw estimate, confidence interval, and bounds are on the original
## scale; the Hedges's g and its bounds are relative to mu = 7.5.
```

The same interface handles paired designs by setting `paired = TRUE`. SMDs are reported automatically; Hedges's correction is applied by default but can be toggled with `bias_correction`, and Glass's delta can be requested via the `glass` argument. Setting equivalence bounds in SMD units (via `eqbound_type = "SMD"`) is possible but discouraged because the bound becomes dependent on sample variance (see the Standardized Effect Sizes section below).

3.3. *Checking Assumptions*

The parametric t-tests implemented in `t_TOST` assume independent observations, approximately normal residuals (or differences, in the paired and one-sample cases), and, for Student's form, equal variances between groups; Welch's correction relaxes the equal-variance assumption and is the default. The residual normality assumption matters most in small samples, where the central limit theorem provides limited protection. Visual diagnostics are generally more informative than formal tests: QQ plots of residuals or differences for normality, side-by-side boxplots or density overlays for variance comparison, and inspection of residual plots or boxplots for extreme or influential observations. A worked diagnostic workflow for each design is provided in the online supplement. When these assumptions are seriously in doubt, the bootstrap and permutation alternatives discussed in later sections provide valid inference under relaxed — though not eliminated — distributional requirements.

3.4. *Using Summary Statistics*

When only summary statistics are available (e.g., from a published article), the `tsum_TOST` function performs the same tests using sample means, standard deviations, sample sizes, and (for paired designs) the correlation between measures[1]:

```
res_tsum = tsum_TOST(
  # sample mean
  m1 = mean(bugs$LDHF, na.rm = TRUE),
  # sample standard deviation
  sd1 = sd(bugs$LDHF, na.rm = TRUE),
  # sample size
  n1 = length(na.omit(bugs$LDHF)),
  # equivalence rather than minimal effects test
  hypothesis = "EQU",
  # sets type of SMD confidence interval
  smd_ci = "t",
  # equivalence bounds
  eqb = c(5.5, 8.5)
)

res_tsum
```

```
##
## One-sample t-test
```

[1]The `extract_r_paired` function can be used if the correlation between paired observations is not readily available.

```
##
## The equivalence test was significant, t(90) = -4.244, p = 2.66e-05
## The null hypothesis test was significant, t(90) = 27.942, p = 3.91e-46
## NHST: reject null significance hypothesis that the effect is equal to zero
## TOST: reject null equivalence hypothesis
##
## TOST Results
##                 t df p.value
## t-test     27.942 90 < 0.001
## TOST Lower  7.116 90 < 0.001
## TOST Upper -4.244 90 < 0.001
##
## Effect Sizes
##            Estimate     SE             C.I. Conf. Level
## Raw           7.379 0.2641  [6.9402, 7.818]         0.9
## Hedges's g    2.905 0.2395 [2.4289, 3.3804]         0.9
## Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
## Equivalence Bounds: Raw [5.5, 8.5]; Hedges's g [2.1832, 3.3741]
```

4. Simplified Hypothesis Testing

The `t_TOST` function is designed to be comprehensive in that it runs three tests, computes standardized effect sizes, and produces verbose output. This is useful for exploration or situations where researchers *always* want to pair a traditional nil significance test with an equivalence test. However, sometimes a researcher has a single specific hypothesis to test, or simply wants less verbose output (e.g., SMD calculation is not wanted). The `simple_htest` function provides a lighter interface for this purpose. It returns a standard R `htest` object (the same class returned by `t.test()` and `wilcox.test()`) but with expanded `alternative` options that include `"equivalence"` and `"minimal.effect"`. This means researchers can perform equivalence testing with the same familiar interface they already use for standard hypothesis tests[2].

Alongside `simple_htest`, a set of helper functions makes it easy to work with any `htest` object: `as_htest` converts verbose TOST output to the simpler format, `describe_htest` generates formatted text for reporting, `df_htest` converts results to data frames for tabulation, and `plot_htest_est` produces estimate plots with equivalence bounds. Checking assumptions would follow the same workflow as `t_TOST`, as both are using t-tests, as discussed in the online supplement.

4.1. *The simplified function*

The `simple_htest` function supports `t.test` and `wilcox.test` as the underlying test (via the `test` argument) and all five alternative hypotheses: `"two.sided"`, `"less"`, `"greater"`, `"equivalence"`, and `"minimal.effect"`. The `mu` parameter specifies equivalence bounds. This single value creates symmetric bounds (e.g., if 1 is pro-

[2]In fact, users can mostly replace `t.test()` and `wilcox.test()` with `simple_htest` as it also allows for the standard alternative hypotheses of two-sided, greater, less. If you have a hypothesis that would traditionally utilize a t-test or Wilcoxon-Mann-Whitney (WMW) test, then you can jump right to using `simple_htest` in TOSTER

vided at `mu` with `alternative = "equivalence"` then the equivalence region will be [-1,1]), or two values can be given for asymmetric bounds.

```
res_sh1 = simple_htest(
  extra ~ group,
  data = sleep,
  paired = TRUE,
  # equivalence hypothesis
  alternative = "equivalence",
  # symmetric bounds of -0.5 and 0.5
  mu = 0.5
)
res_sh1
```

```
##
##  Paired t-test
##
## data:  extra by group
## t = -2.7766, df = 9, p-value = 0.9892
## alternative hypothesis: equivalence
## null values:
## mean difference mean difference
##            -0.5             0.5
## 90 percent confidence interval:
##  -2.2930053 -0.8669947
## sample estimates:
## mean of the differences (z = '1' - '2')
##                                   -1.58
```

Additionally, tests other that TOST can be applied with this function. The traditional nil-hypothesis tests, non-inferiority, or even superiority-by-a-margin hypotheses can all be tested with this function. For example, let's say we want to test that a treatment group in the sleep data was just non-inferior to the other treatment rather than equivalent. This can be accomplished by just setting the direction with the `alternative` argument and the margin of non-inferiority by the `mu` argument.

```
res_sh2 = simple_htest(
  extra ~ group,
  data = sleep,
  paired = TRUE,
  # non-inferiority rather than equivalence
  alternative = "greater",
  # margin of non-inferiority
  mu = -0.5
)
res_sh2
```

```
##
##  Paired t-test
##
## data:  extra by group
```

```
## t = -2.7766, df = 9, p-value = 0.9892
## alternative hypothesis: true mean difference is greater than -0.5
## 95 percent confidence interval:
##  -2.293005       Inf
## sample estimates:
## mean of the differences (z = '1' - '2')
##                                   -1.58
```

Notice how the output, in either hypothesis test above, is a single clean `htest` object. This is much more concise than the verbose `t_TOST` output shown previously on purpose. If you want to perform additional tests (such as a complementary nil significance tests) then that has to be done separately. However, there are some methods (described below) that make this `htest` format easy to deal with when reporting statistical test results.

4.2. *Describing and Tabulating Results*

For writing up results, `describe_htest` generates a formatted text string following typical scientific reporting conventions:

```
cat(describe_htest(res_sh1))
```

The Paired t-test is not statistically significant (t(9) = -2.78, p = 0.989, mean of the differences (z = '1' - '2') = -1.58, 90% C.I.[-2.29, -0.867]) at a 0.05 alpha-level. The null hypothesis cannot be rejected. At the desired error rate, it cannot be stated that the true mean difference is between -0.5 and 0.5.

4.3. *Plotting Results*

A visual summary of the estimate and confidence interval can be produced with `plot_htest_est`. For equivalence and minimal effect tests, the equivalence bounds are shown as dashed vertical lines:

```
plot_htest_est(res_sh1)
```

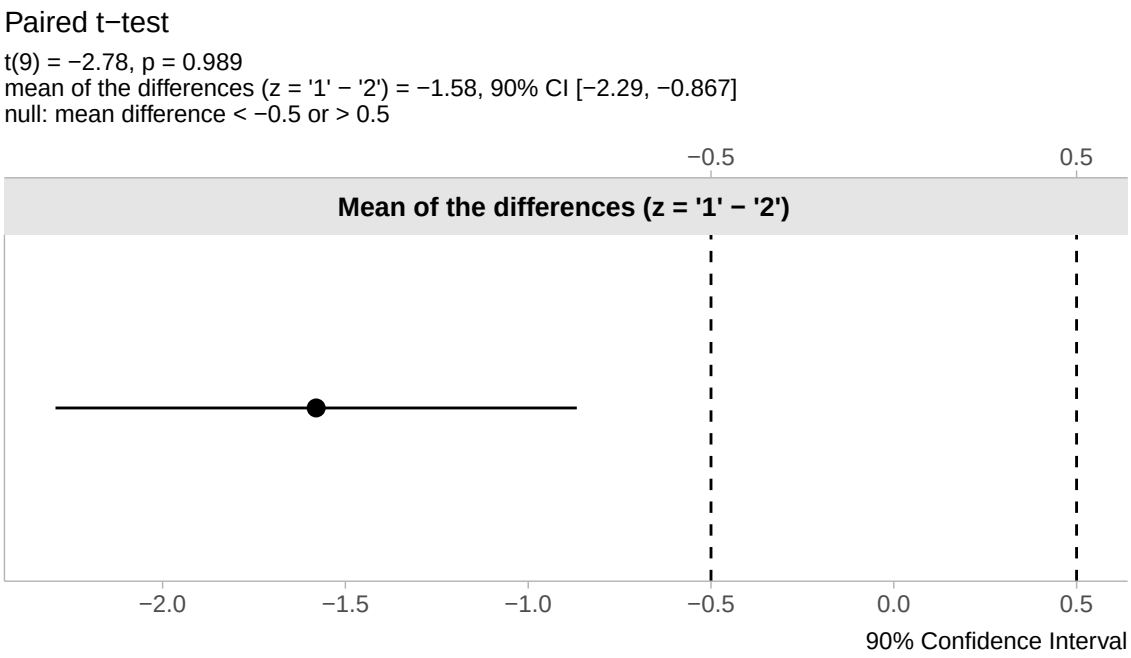


Figure 4. Estimate plot for an equivalence test via simple_htest.

5. Standardized Effect Sizes

This section provides a unified treatment of the standardized effect size tools in TOSTER and I cover SMDs, correlation coefficients, and tools for comparing both between studies. Before demonstrating these tools, it is worth discussing the limitations of standardized effect sizes, because these limitations are especially important for equivalence testing.

5.1. *A Brief Warning on Standardized Effect Sizes*

Standardized effect sizes — Cohen's d, correlation coefficients, standardized regression coefficients — are appealing because they allow comparison across studies and scales, are required by many journals, and serve as the primary currency of meta-analysis. However, they have well-documented problems that are especially relevant to equivalence testing.

SMDs are sensitive to the sample variance. The "same" real-world effect produces different d values across populations and study designs. The ubiquitously reported Cohen's d conflates the magnitude of the mean difference with the dispersion of the observed outcome, which means that a narrow equivalence bound on d may correspond to a very different raw-scale bound depending on the variability of the sample (Caldwell and Vigotsky 2020; Baguley 2009). This makes it unclear what, substantively, is being declared "equivalent". Are we claiming the means are close, or that the means are close relative to the variance? The choice of denominator (pooled SD, control SD, within-subject SD) further complicates matters.

Correlations have analogous issues. Tukey (1969) noted that correlations are influenced by the range of both variables, so the "same" bivariate relationship pro-

duces different coefficients depending on sample selection. A correlation of 0.3 in a restricted-range sample may correspond to a much larger association in the full population. Standardized regression coefficients share these vulnerabilities. These depend on the variances of both the predictor and outcome, which makes them unstable across samples and difficult to interpret substantively (Greenland et al. 1991).

The implication for equivalence testing is straightforward. When the measurement scale is meaningful, researchers should generally prefer *unstandardized* (raw-scale) equivalence bounds. When standardized bounds are used, the choice should be justified with reference to the specific population and measurement context, not borrowed from generic benchmarks.

In fairness to those who prefer standardization, it can seem prudent in some situations (e.g., meta-analysis), though in an original study a better-suited estimand is usually available. TThere are two common situations where I think researchers reach for an SMD when better solutions exist. When the standard deviation is proportional to the mean, a ratio of means on the log scale captures a stable and interpretable quantity (see Log-Transformed Methods), and when an outcome lacks meaningful units, tests of stochastic superiority offer an easy-to-interpet alternative (see Brunner-Munzel Test). As far as I can tell, standardization is most defensible when the standardizer is fixed externally (e.g., a normative population SD; which essentially means it is a amounts to a raw-scale bound), or when comparing studies that used different instruments. Even then, differences in standardization can manufacture apparent heterogeneity that only reflect design, study population, and analytic choices.

With these caveats in mind, the following subsections describe the tools TOSTER provides for computing and testing standardized effect sizes assuming they will be utilized an appropriate amount of skepticism.

5.2. *Standardized Mean Difference Functions*

The `smd_calc` function computes an SMD — Cohen's d, Hedges' g (bias-corrected), or Glass's Δ — with confidence intervals and *optional* hypothesis testing (utilizing a z-test: $z = (d - \delta_0)/SE_d$, where d is the sample SMD estimate, δ_0 is the null value being tested, and SE_d is its standard error.). It supports one-sample, paired, and two-sample designs, and can return either an `htest` object or a data frame. The SMD is already reported in `t_TOST` output, but `smd_calc` allows standalone computation and supports equivalence testing directly on the effect size. This may be useful as a complementary function when using `simple_htest` as the primary statistical testing function.

When estimation rather than testing is the goal, `smd_calc` can be called with `alternative = "none"` (the default), which returns only the point estimate and confidence interval without performing a hypothesis test. This is the recommended usage when the primary objective is to report the magnitude of the effect.

```
smd_calc(
  formula = extra ~ group,
  data = sleep,
  paired = TRUE,
  # apply Hedges' correction
  bias_correction = TRUE,
  # apply the repeated measures correction
  rm_correction = TRUE,
```

```
  # return an htest object rather than a data frame
  output = "htest"
)
```

```
##
##  Paired Sample Standardized Mean Difference (SMD; Hedges's
##  g[rm]=('1'-'2')/SD_rm)
##
## data:  extra by group
##
## alternative hypothesis: none
## 95 percent confidence interval:
##  -1.39802439 -0.07479266
## sample estimates:
## SMD (g[rm])
##  -0.7513666
```

When hypothesis testing on the SMD is desired, `smd_calc` supports equivalence and minimal-effect alternatives directly:

```
smd_calc(
  formula = extra ~ group,
  data = sleep,
  paired = TRUE,
  # apply Hedges' correction
  bias_correction = TRUE,
  # apply the repeated measures correction
  rm_correction = TRUE,
  # equivalence hypothesis
  alternative = "equivalence",
  # equivalence bounds on the SMD
  null.value = c(-0.5, 0.5),
  # return an htest object rather than a data frame
  output = "htest"
)
```

```
##
##  Paired Sample Standardized Mean Difference (SMD; Hedges's
##  g[rm]=('1'-'2')/SD_rm)
##
## data:  extra by group
## z = -0.78495, p-value = 0.7838
## alternative hypothesis: equivalence
## null values:
## lower bound upper bound
##        -0.5         0.5
## 90 percent confidence interval:
##  -1.2894564 -0.1789204
## sample estimates:
## SMD (g[rm])
```

```
##  -0.7513666
```

However, the parametric z-test used by `smd_calc` relies on distributional assumptions about the sampling distribution of the SMD that may not hold in practice. When a hypothesis test on the SMD is the goal, bootstrapping via `boot_smd_calc` is recommended because it provides more robust confidence intervals and p-values[3].

```
boot_smd_calc(
  formula = extra ~ group,
  data = sleep,
  paired = TRUE,
  # apply Hedges' correction
  bias_correction = TRUE,
  # apply the repeated measures correction
  rm_correction = TRUE,
  # equivalence hypothesis
  alternative = "equivalence",
  # equivalence bounds on the SMD
  null.value = c(-0.5, 0.5)
)
```

```
##
##  Paired Sample bootstrapped Standardized Mean Difference (SMD; Hedges's
##  g[rm]=('1'-'2')/SD_rm)
##
## data:  extra by group
## z-observed = -0.78495, p-value = 0.8956
## alternative hypothesis: equivalence
## null values:
## lower bound upper bound
##        -0.5         0.5
## 90 percent confidence interval:
##  -1.1748220 -0.4246855
## sample estimates:
## SMD (g[rm])
##  -0.7513666
```

5.2.1. Checking Assumptions

The SMD functions inherit the assumptions of the underlying t-test and therefore the visual diagnostics from the t-test discussion above apply directly. However, two considerations are specific to standardized effect sizes. First, outliers and skew affect both the numerator (mean difference) and the denominator (standard deviation) of the SMD, which can amplify their influence on the estimate; the `tr` trimmed-means argument available in the `boot_smd_calc` function is particularly worth considering when the raw data show extreme values. Second, the asymptotic z-test in `smd_calc` relies on an approximation to the sampling distribution of the SMD that can be very inaccurate

[3] In a limited set of simulations, I found that only the BCa bootstrap maintained near-nominal coverage under substantial skewness with small samples ($n < 50$). I am currently developing refinements to the studentized bootstrap that should improve its coverage in these settings as well.

in small samples, which is the primary motivation for preferring `boot_smd_calc` when hypothesis testing is the goal.

5.3. *Equivalence Tests for Correlations*

Researchers sometimes need to test that a correlation is practically zero. The `z_cor_test` function provides a asymptotic approach using the Fisher z-transformation. However, this is only an approximate method and is generally not recommended as the primary inferential tool for equivalence testing on correlations.

```
z_cor_test(
  x = bugs$LDLF,
  y = bugs$LDHF,
  # type of correlation coefficient
  method = "pearson",
  # two-sided nil hypothesis test
  alternative = "t"
)
```

```
##
##  Pearson's product-moment correlation with approximate SE
##
## data:  bugs$LDLF and bugs$LDHF
## z = 6.169, N = 91, p-value = 6.871e-10
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.4208204 0.6996192
## sample estimates:
##         r
## 0.5767783
```

Because the z-test relies on distributional approximations that can be inaccurate — particularly with small samples — bootstrapping via `boot_cor_test` is recommended for more robust inference. The `boot_cor_test` function supports all three major correlation methods (Pearson, Kendall, and Spearman) as well as the Winsorized (Wilcox 1993) and percentage bend (Wilcox 1994) correlation coefficients as robust alternatives, and returns standard `htest` objects that integrate with the helper functions described above.

```
res_boot_cor = boot_cor_test(
  x = bugs$LDLF,
  y = bugs$LDHF,
  # type of correlation coefficient
  method = "pearson",
  # equivalence hypothesis
  alternative = "equivalence",
  # equivalence bounds of -0.4 and 0.4
  null = 0.4
)
```

```r
print(res_boot_cor)
```

```
##
##  Bootstrapped Pearson's product-moment correlation (studentized)
##
## data:  bugs$LDLF and bugs$LDHF
## N = 91, p-value = 0.9795
## alternative hypothesis: equivalence
## null values:
## correlation correlation
##         0.4        -0.4
## 90 percent confidence interval:
##  0.4364097 0.6886763
## sample estimates:
##         r
## 0.5767783
```

```r
cat(describe_htest(res_boot_cor))
```

The Bootstrapped Pearson's product-moment correlation (studentized) is not statistically significant (p = 0.979, r = 0.577, 90% C.I.[0.436, 0.689]) at a 0.05 alpha-level. The null hypothesis cannot be rejected. At the desired error rate, it cannot be stated that the true correlation is between 0.4 and -0.4.

If only summary statistics are available, an approximate method is provided through `corsum_test`, which uses the Fisher z-transformation to compute confidence intervals and p-values.

5.3.1. Checking Assumptions

Correlation-based inference in `z_cor_test` and `boot_cor_test` assumes independent paired observations and, for Pearson's coefficient, a roughly linear relationship between variables with approximately symmetric marginal distributions; Spearman and Kendall relax the linearity requirement to monotonicity, making them appropriate when the relationship is non-linear but directionally consistent. The bootstrap version in `boot_cor_test` further relaxes distributional assumptions about the marginals but remains sensitive to observations that exert disproportionate influence on the estimate. Visual diagnostics are the primary tools: scatterplots with a loess smoother to assess linearity or monotonicity, marginal density plots or QQ plots to check distributional shape, and jackknife-after-bootstrap plots to identify individual observations driving bootstrap variability. A worked diagnostic workflow is provided in the online supplement.

5.4. *Comparing Standardized Effect Sizes Between Studies*

When evaluating whether a replication study produced results consistent with the original, researchers need a way to formally compare effect sizes. The `compare_smd` function tests whether two SMD estimates (e.g., original study compared to replication study) differ more than would be expected by sampling variability alone.

Consider an original study reporting Cohen's $d_z = 0.95$ with 25 participants and a replication with $d = 0.23$ and 50 participants. A researcher might wonder if these

these estimates compatible, and it would be prudent to formally test this. Let's say these researchers determine that equivalence between studies would be SMDs within 0.25 of each other. We can then perform a significance and equivalence test between the two original and replication study estimates.

```
# Significance test
compare_smd(
  # SMD and sample size of the original study
  smd1 = 0.95,
  n1 = 25,
  # SMD and sample size of the replication
  smd2 = 0.23,
  n2 = 50,
  # both estimates come from paired designs
  paired = TRUE,
  # nil hypothesis test
  null = 0,
  # significance hypothesis
  alternative = "two.sided"
)
```

```
##
##  Difference in Cohen's dz (paired)
##
## data:  Summary Statistics
## z = 2.5685, p-value = 0.01021
## alternative hypothesis: true difference in SMDs is not equal to 0
## sample estimates:
## difference in SMDs
##               0.72
```

```
# Equivalence test
compare_smd(
  # SMD and sample size of the original study
  smd1 = 0.95,
  n1 = 25,
  # SMD and sample size of the replication
  smd2 = 0.23,
  n2 = 50,
  # both estimates come from paired designs
  paired = TRUE,
  # equivalence bounds on the difference in SMDs
  null = .25,
  # equivalence hypothesis
  alternative = "equivalence"
)
```

```
##
##  Difference in Cohen's dz (paired)
##
```

```
## data:  Summary Statistics
## z = 1.6767, p-value = 0.9532
## alternative hypothesis: equivalence
## null values:
## difference in SMDs difference in SMDs
##               0.25              -0.25
## sample estimates:
## difference in SMDs
##               0.72
```

Based on these results, we would reject the null significance hypothesis (conclude the SMDs differ) but fail to reject the equivalence null hypothesis (we cannot conclude the SMDs are practically equivalent). When raw data are available, `boot_compare_smd` provides a bootstrap alternative (see the package vignettes for details).

For comparing correlation coefficients between independent studies, the `compare_cor` function provides an analogous test. It uses either the Fisher z-transformation (default) or the Kraatz (Anderson-Hauck) method to compare two correlation coefficients.

```
compare_cor(
  # correlation and degrees of freedom of the first study
  r1 = 0.45,
  df1 = 48,
  # correlation and degrees of freedom of the second study
  r2 = 0.25,
  df2 = 78,
  # equivalence hypothesis
  alternative = "equivalence",
  # equivalence bounds on the difference in correlations
  null = 0.25
)
```

```
##
##  Difference between two independent correlations (Fisher's z transform)
##
## data:  Summary Statistics
## z = -0.14114, p-value = 0.4439
## alternative hypothesis: equivalence
## null values:
## difference between correlations difference between correlations
##                            0.25                           -0.25
## sample estimates:
## difference between correlations
##                             0.2
```

When raw data are available for both studies, `boot_compare_cor` provides a bootstrap alternative that avoids the distributional approximations underlying the Fisher z and Kraatz methods.

The comparison functions `compare_smd` and `compare_cor` inherit the assumptions of the underlying effect size estimates they combine. Because both studies contribute to the comparison, violations in either study propagate into the test. When raw data

are available, the bootstrap alternatives (`boot_compare_smd` and `boot_compare_cor`) are preferred because they avoid the Fisher z and related asymptotic approximations, which can be inaccurate particularly when effect sizes are large or sample sizes are modest. When only summary statistics are available, readers should report the diagnostic context of the original and replication studies where possible.

6. Robust Methods for Equivalence Testing

When t-test assumptions (normality, homoscedasticity) are seriously violated, or when the mean difference is a poor summary of the effect (e.g., with heavy skew or ordinal psychometric scales) alternative approaches are worth considering. I have organized TOSTER into three robust alternative categories: rank-based tests, resampling methods, and log-transformed methods.

6.1. *Rank-Based Tests*

Rank-based tests operate on the ranks of the data rather than the raw values, which is why they are typically described as non-parametric. It is worth emphasizing that this label applies to the test, not the data. The underlying observations may still come from any distribution, and rank-based methods should not be treated as automatic substitutes for a t-test when normality is in doubt. These tests are not tests of means or medians (Divine et al. 2018; Karch 2021); they target different estimands (i.e., the quantity a study is attempting to estimate), and the choice between a rank-based test and a t-test should be driven by which estimand is of scientific interest.

6.1.1. Wilcoxon-Mann-Whitney Tests

The Wilcoxon-Mann-Whitney (WMW) family is the rank-sum test for two independent groups and the signed-rank test for paired and one-sample designs. It provides non-parametric tests based on ranks. Within the TOST framework, TOSTER performs two directional tests to determine whether the effect lies within (equivalence) or outside (minimal effect) the equivalence bounds. What that effect is depends on an assumption that researchers often do not realize they are making, and there are persistent misconceptions worth addressing directly. The three subsections below separate what the WMW tests evaluate with no additional assumptions, what they evaluate under a location-shift assumption, and what `wilcox_TOST` actually places bounds on.

6.1.1.1. Without a location-shift assumption: a test of stochastic equality. Many researchers believe the WMW tests are tests of medians. They are not. With no assumption beyond independence and exchangeability, the WMW test evaluates stochastic equality which has the null hypothesis that a randomly drawn observation from one group is equally likely to exceed, or to be exceeded by, a randomly drawn observation from the other,

$$H_0 : P(X > Y) + \frac{1}{2} \cdot P(X = Y) = 0.5$$

As Fay and Malinovsky (2018) demonstrate, the effect size accompanying the test is therefore most naturally expressed as stochastic superiority (or dominance) rather than as a difference in central tendency. Nothing about this null hypothesis refers to a mean or a median.

6.1.1.2. Under a location-shift assumption: a test of central tendency. The location-shift assumption holds that the two distributions are identical in every respect except location. Under this assumption, and only under it, stochastic equality

and equality of medians coincide, and the Hodges-Lehmann estimator can be read as an estimate of the median difference. This is the interpretation most readers have in mind when they reach for a WMW test.

When the assumption fails, the two interpretations can come apart dramatically. Divine et al. (2018) demonstrate striking counterexamples: groups can have equal medians yet produce a significant WMW test, very different medians yet a non-significant WMW test, or even significance in the direction opposite the median comparison. Critically, nothing in the test output signals that the assumption has failed, and the p-value and the Hodges-Lehmann estimate are reported the same way regardless. The diagnostic described below is therefore not optional if a central-tendency interpretation is intended.

6.1.1.3. What `wilcox_TOST` places bounds on. In TOSTER, `wilcox_TOST` applies the equivalence bounds to the pseudo-median, which corresponds to the mean or median difference only under the location-shift assumption. This has a direct practical consequence for setting bounds: if you are not prepared to assert location shift, the pseudo-median is your estimand, and the smallest effect size of interest must be justified on that scale rather than imported from reasoning about mean differences.

If the hypothesis genuinely concerns means, `perm_t_test` or `boot_t_test` are the more appropriate tools. If it concerns stochastic superiority, `brunner_munzel` provides a clearer framework and a directly interpretable effect size.

```
test1 = wilcox_TOST(
  formula = extra ~ group,
  data = sleep,
  paired = TRUE,
  # equivalence bounds on the pseudo-median
  eqb = .5
)
print(test1)
```

```
##
## Wilcoxon signed rank test with continuity correction
##
## The equivalence test was non-significant V = 0.000, p = 9.96e-01
## The null hypothesis test was significant V = 0.000, p = 9.09e-03
## NHST: reject null significance hypothesis that the effect is equal to zero
## TOST: don't reject null equivalence hypothesis
##
## TOST Results
##            Test Statistic p.value
## NHST                    0   0.009
## TOST Lower              2   0.996
## TOST Upper              0   0.003
##
## Effect Sizes
##                            Estimate          C.I. Conf. Level
## Median of Differences          -1.4 [-2.7, -1.15]          0.9
## Rank-Biserial Correlation      -1.0 [-1, -0.4492]          0.9
```

Based on these results, we would conclude there is a significant difference and the results are also not equivalent. But again, the estimate we are testing is the pseudo-median, not the mean or median difference, so this conclusion may not be relevant to the research question at hand.

6.1.1.4. Checking Assumptions. The WMW tests assume independent observations and exchangeability under the null, the latter meaning that group labels carry no information about outcomes if there is no effect. Those two assumptions are all that the stochastic-equality interpretation requires. The two assumptions below determine whether the stronger central-tendency interpretation is available, and each has a straightforward visual diagnostic.

The location-shift assumption is that the two group distributions differ only in location, not in shape or spread. This is what allows WMW results to be interpreted as a test of the median difference via the Hodges-Lehmann estimator. It is best diagnosed by overlaying the empirical cumulative distribution functions (ECDFs) of the two groups. Under location shift, the curves are horizontal translations of one another, so the horizontal distance between them is constant across the range of the data (Figure 5A). Departures appear as a gap that widens, narrows, or reverses sign, and in the clearest cases as curves that cross (Figure 5B). In panel B, the two groups there differ in both location and spread, so there is no single number that describes "the shift," and the Hodges-Lehmann estimate that `wilcox_TOST` reports does not correspond to a difference in medians.

For the signed-rank variants used by `wilcox_TOST` and `simple_htest`, the difference distribution must additionally be symmetric about the null value for the pseudo-median to coincide with the mean or median difference. Asymmetry is readily checked with a symmetry plot of the paired differences (Figure 6). As with the location-shift case, visible departures suggest that the test's output (i.e., the pseudo-median and associated p-value) does not map cleanly onto a hypothesis about central tendency.

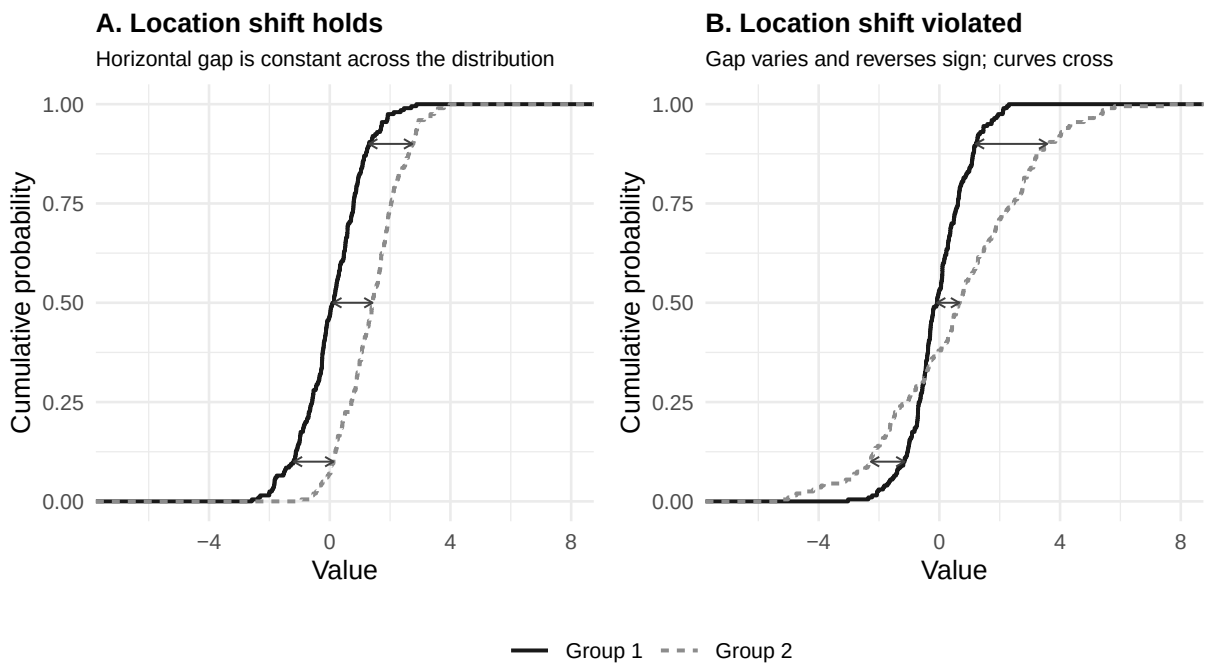


Figure 5. Empirical cumulative distribution functions for two simulated groups, with double-headed arrows marking the horizontal distance between the curves at the 10th, 50th, and 90th percentiles. (A) The location-shift assumption holds: the distributions differ only in location, the arrows are of equal length, and the Hodges-Lehmann estimator can be read as a median difference. (B) The assumption is violated: the groups differ in spread as well as location, so the horizontal gap varies in magnitude and reverses sign across the distribution, and the curves cross. In panel B there is no single shift to estimate, and a central-tendency interpretation of the WMW test is not available.

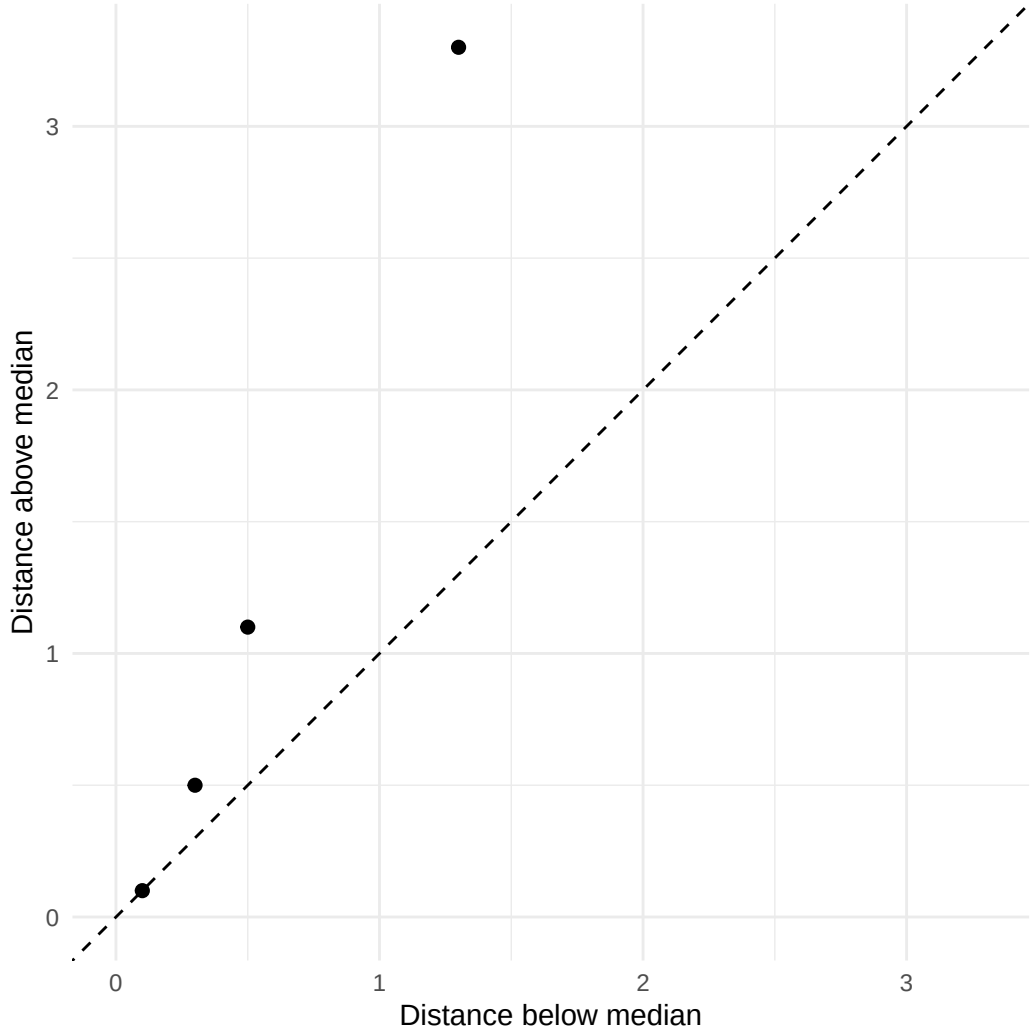


Figure 6. Symmetry plot of the paired differences from the sleep dataset, pairing each observation's distance above the median with the corresponding distance below. Under symmetry, points fall along the 45-degree reference line (dashed); systematic departures — points consistently above or below the line — indicate asymmetry. The signed-rank variant of WMW requires this symmetry for the pseudo-median to coincide with the mean or median of the differences.

6.1.2. Brunner-Munzel Test

The Brunner-Munzel test (Brunner and Munzel 2000) estimates the relative effect $\hat{p} = P(X > Y) + 0.5 \cdot P(X = Y)$, sometimes referred to as the probability of superiority or stochastic superiority. As mentioned in the WMW section, this is the probability that a randomly chosen observation from one group exceeds a randomly chosen observation from the other, with ties split evenly. This estimand is interpretable without distributional assumptions, unlike the WMW test which requires a location-shift assumption to say anything about location differences. Karch (2021) argues it should be preferred as the default non-parametric procedure. Its main advantage over WMW is that the null hypothesis of stochastic equality does not require the two distributions to be identical (i.e., the nonparametric Behrens–Fisher problem). WMW's Type I error rate can depart from nominal when the groups are stochastically equal but differ in variance or shape; the Brunner–Munzel test remains asymptotically valid in that setting and provides a directly interpretable effect size.

Stochastic superiority tests, like the Brunner-Munzel test, may be particularly well-suited to Likert-type scales which are ubiquitous in psychological research. The spacing between items on an ordinal scale has no guaranteed meaning, and a mean difference therefore has a strained interpretation. For example, what, concretely, does a 0.4-unit mean difference on a five-point Likert-type scale communicate? An estimate of stochastic superiority sidesteps this interpretation difficulty. The probability that a randomly selected participant from one group outscores a randomly selected participant from the other is interpretable on its own terms, with no knowledge of the scale required. I am not suggesting that using t-tests will always mislead in these contexts but it is a concern that should not be quickly dismissed either (Liddell and Kruschke 2018). Additionally, even if we assume traditional parametric methods are robust to the ordinal nature of the data, the preserved error rate is not the same as interpretive clarity. For researchers whose primary outcomes live on ordinal scales, the Brunner-Munzel test and its associated effect size — as well as ordinal models more generally (Bürkner and Vuorre 2019) — may offer a more natural and defensible statistical analysis option.

The `brunner_munzel` function supports equivalence and minimal effect testing via `alternative`, with bounds on the probability scale via `mu`. Results can be reported on different scales (`"probability"`, `"difference"`, `"logodds"`, or `"odds"`), and four test methods are available: `"t"` (default), `"logit"` (range-preserving CIs)`"perm"` (studentized permutation; Neubert and Brunner (2007)), and `"perm_logit"` (studentized logit-transformed permutation; borrows advantages of both logit and permutation approach). Below, the test is applied to the same female and male `LDHF` ratings used in the independent-groups t-test example, which, as bounded rating-scale responses, are a natural candidate for this estimand.

```
set.seed(506119)
bm_test = brunner_munzel(
  formula = LDHF ~ Gender,
  data = bugs_g,
  # equivalence hypothesis
  alternative = "equivalence",
  # equivalence bounds
  # on the stochastic superiority scale
  mu = c(0.3, 0.7),
  # Permutation logit
```

```r
  # Best performance (so far) for equivalence testing
  test_method = "perm_logit"
)
print(bm_test)
```

```
##
##  Two-sample Brunner-Munzel Randomization test (logit)
##
## data:  LDHF by Gender
## t-observed = -2.0766, N-permutations = 10000, p-value = 0.0174
## alternative hypothesis: equivalence
## null values:
## lower bound upper bound
##         0.3         0.7
## 90 percent confidence interval:
##  0.4763624 0.6742033
## sample estimates:
## P(Female>Male) + .5*P(Female=Male)
##                          0.5785755
```

6.1.2.1. Checking Assumptions. The Brunner-Munzel test has few assumption requirements. It requires independent observations within and between groups, and (for the t-approximation used by default) sample sizes sufficient for the asymptotic approximation to be reasonable. Unlike proportional-odds models, the constant-odds-ratio assumption is not required. The relative effect estimated within the Brunner-Munzel test is defined directly from the two marginal distributions and carries the same interpretation — the probability that a random observation from one group exceeds one from the other — regardless of the shapes of the distributions. However, I would still recommend visual inspection of the data and their distribution since it is useful for understanding the data, but there is no distributional assumption that these checks are verifying. Additionally, I would strongly recommend using the studentized logit-tranformed permutation variant (`method = "perm_logit"`) for equivalence testing, and especially with small samples, as simulations have indicated this method is the most powerful while also preserving the type 1 error rate.

6.2. *Resampling Methods*

When the goal is to test hypotheses about means, resampling methods are powerful tools that allow distributional assumptions to be relaxed. TOSTER implements both a bootstrap and a permutation approach through the `boot_t_test` and the `perm_t_test`, respectively. The methods might slightly differ in inferential logic but both evaluate the same estimand (the difference in means) as the Student's t-test, but with fewer distributional constraints.

The test statistic itself is unchanged for either the bootstrap or permutation t-test. The only thing that changes is how the null distribution is handled. In the conventional t-test, the null distribution of the mean difference is derived analytically under the assumption that the data are drawn from a normal distribution, with equal variances assumed in Student's t-test and unequal variances accommodated in Welch's t-test. The bootstrap approximates the sampling distribution of the same statistic by resampling from the observed data with replacement, whereas the permutation approach constructs an exact or near-exact null distribution (under the sharp null hypothesis of identical distributions) by shuffling group labels (two-sample) or flipping signs (one-sample and paired). In both cases, the researcher continues to ask the same question about means; only the calibration of the reference distribution differs.

Distributional requirements are relaxed, though not eliminated, because the reference distribution is generated from the data (rather than making assumptions about the distribution). The bootstrap relies on asymptotic arguments and assumes that the empirical distribution is a reasonable stand-in for the population distribution. It can perform well with moderate-to-large samples, but can struggle with very small samples, heavy tails, or extreme outliers unless trimmed means are used. The permutation approach is exact under the null hypothesis of exchangeability, meaning group labels carry no information about outcomes. In its basic form this exactness requires identical distributions under the null, which implicitly assumes equal variances. The studentized permutation test (Janssen 1997; Chung and Romano 2013) addresses this limitation by permuting a scale-invariant statistic (essentially the Welch t-statistic), yielding asymptotically valid inference even under heteroscedasticity. Arboretti, Pesarin, and Salmaso (2020) developed permutation approaches to equivalence testing; TOSTER's implementation is a simpler, uncalibrated variant that is asymptotically equivalent under studentization (though less powerful). In short, these methods are assumption-lighter rather than assumption-free. I would strongly recommend that users still consider whether these assumptions are reasonable for their design.

Also, as a quick word of caution, a common reflex when normality seems implausible is to switch to the WMW or other rank-based tests. However, as stated throughout this manuscript, these tests answer a different question and offer a different estimate. As discussed earlier in this manuscript, rank-based procedures test hypotheses about stochastic ordering or pseudo-medians rather than means. When the scientific question is genuinely about means — as it often is in applied research where effect sizes and interventions are naturally expressed on the original scale — bootstrap and permutation t-tests should generally be preferred over rank-based methods for handling such violations of assumptions. The rank-based tests are useful, but as answers to different questions, and should never be considered as robust substitutes for tests of means.

The choice of the bootstrap or the permutation t-test involves weighing the trade-offs between tests. Both are tests of means, and both return confidence intervals that are consistent with their p-values. The internal mechanisms of each type of t-test inform their usefulness. In a randomized experiment the permutation distribution is the

randomization distribution which makes permutation tests attractive for randomized experiments. However, the permutation test is only exact under the sharp null hypothesis that the two groups have identical distributions, at any sample size. When studentized, it is also asymptotically valid under the weak null hypothesis that only the means are equal (Janssen 1997; Chung and Romano 2013). The studentized permutation test has the strongest theoretical footing for two-sample mean comparisons under heteroscedasticity, and is particularly attractive for equivalence testing (Arboretti, Pesarin, and Salmaso 2020). On the other hand, the bootstrap only targets the weak null, and is only asymptotically valid (never exact). In general, the use of bootstrap methods offers greater flexibility because it extends naturally to statistics beyond the mean difference, provides readily interpretable confidence intervals via CI inversion (Thulin 2024). However, for both methods validity under the weak null is only asymptotic. In very small sample sizes ($n < 10$) with unequal variances (1:4), even these robust methods can be somewhat liberal (e.g., type 1 error rising to around 8% at nominal alpha of 0.05). When outliers are a concern, both methods accept the `tr` argument for trimmed means, which reduces the influence of extreme observations. Both are substantial improvements over ignoring assumption violations of the t-test, and both return standard `htest` objects that integrate with the TOST framework for equivalence testing. Within the package documentation, there is some rough guidance on which method to prefer depending on various simulated scenarios. As a rough rule, the permutation t-test performed well for two-independent groups, the bootstrap t-test using the studentized bootstrap performed well for paired data. In the case of heavy tails and outliers, the trimming option for either method was very helpful in maintaining the type 1 error rate.

6.2.1. Bootstrap t-test

The bootstrap t-test (`boot_t_test`) in TOSTER defaults to the studentized bootstrap (`boot_ci = "stud"`) because this matches the original bootstrap t-test proposed by Efron and Tibshirani (1993) (Chapter 16), which is the procedure `boot_t_TOST` implements. In this approach, the t-statistic itself is resampled rather than the raw mean difference, and confidence intervals are constructed from the bootstrap distribution of these pivotal quantities. Other CI methods (percentile, basic, and BCa) remain available via the `boot_ci` argument, but the studentized default preserves consistency with the Efron and Tibshirani (1993) formulation on which this function was based upon. It also ensures that the reported p-value and confidence interval are derived from the same bootstrap distribution, so the two will always agree[4].

```
set.seed(4522)
boot_t_test(
  formula = extra ~ group,
  data = sleep,
  paired = TRUE,
  # equivalence hypothesis
  alternative = "equivalence",
  # equivalence bounds
  mu = c(-0.5, 0.5),
  # number of bootstrap resamples
```

[4]Previous versions of the package used the studentized p-value regardless of the selected CI method, which could produce disagreement between reported intervals and p-values.

```
  R = 999
)
```

```
##
##  Bootstrapped Paired t-test (studentized)
##
## data:  extra by group
## t-observed = -2.7766, df = 9, p-value = 1
## alternative hypothesis: equivalence
## null values:
## mean difference mean difference
##            -0.5             0.5
## 90 percent confidence interval:
##  -2.853166 -1.071659
## sample estimates:
## mean of the differences (z = '1' - '2')
##                                   -1.58
```

The output of `boot_t_test` follows the familiar `htest` format. The reported p-value is the larger of the two one-sided tests against the equivalence bounds, and the confidence interval is the studentized bootstrap CI for the mean difference. Because the p-value is tied to this interval via CI inversion, a researcher reporting both will find the two tell a consistent story. The interval will fall within the equivalence bounds if and only if the equivalence test rejects. For situations where both a nil hypothesis test and an equivalence test are desired simultaneously with bootstrapping, `boot_t_TOST` returns a `TOSTt` object with bootstrapped SMD estimates:

```
set.seed(891111)
test1 = boot_t_TOST(
  formula = extra ~ group,
  data = sleep,
  paired = TRUE,
  # equivalence bounds
  eqb = .5,
  # number of bootstrap resamples
  R = 999
)
print(test1)
```

```
##
## Bootstrapped Paired t-test
##
## The equivalence test was non-significant, t(9) = -2.777, p = 9.99e-01
## The null hypothesis test was significant, t(9) = -4.062, p = 0e+00
## NHST: reject null significance hypothesis that the effect is equal to zero
## TOST: don't reject null equivalence hypothesis
##
## TOST Results
##                 t df p.value
## t-test     -4.062  9 < 0.001
```

```
## TOST Lower -2.777  9   0.999
## TOST Upper -5.348  9 < 0.001
##
## Effect Sizes
##                Estimate     SE                C.I. Conf. Level
## Raw              -1.580 0.3699 [-2.8527, -1.0364]         0.9
## Hedges's g(z)    -1.174 0.6491  [-1.3975, 0.9998]         0.9
## Note: studentized bootstrap ci method utilized.
## Equivalence Bounds: Raw [-0.5, 0.5]; Hedges's g(z) [-0.4065, 0.4065]
```

The bootstrapped SMDs are a useful feature in their own right. Analytic confidence intervals for standardized effect sizes rely on approximations, which assume normality and can behave poorly with small or skewed samples. Bootstrap intervals for SMDs sidestep these assumptions by resampling the full estimation pipeline, yielding intervals that inherit the robustness properties of the bootstrap itself (Kirby and Gerlanc 2013).

6.2.1.1. Checking Assumptions. The bootstrap t-test makes fewer distributional assumptions than its conventional Student's t-test but is not assumption-free. It requires independent observations and assumes the empirical distribution is a reasonable stand-in for the population distribution. The latter is an assumption that holds better in moderate-to-large samples than in small ones. The studentized bootstrap is reasonably robust to skewness and heavy tails, but remains sensitive to outliers and other high-influence observations, which can be drawn repeatedly across replicates and disproportionately shape the bootstrap distribution. Useful visual checks include the distribution of the data (histograms, boxplots) to identify extreme values, the bootstrap distribution itself to flag problematic skewness or gaps, and the jackknife-after-bootstrap plot to identify individual observations that dominate the bootstrap variability. The `tr` argument for trimmed means provides a straightforward robustness option when outliers are a concern. Diagnostic workflows for all three designs are demonstrated in the online supplement.

6.2.2. Permutation t-test

When `R = NULL`, the `perm_t_test` function enumerates all possible permutations for an exact test, but otherwise it draws random permutations (size of which is determined by the `R` argument in the function) with the "+1" correction (Phipson and Smyth 2010). By default the test is studentized (`perm_se = TRUE`), consistent with the recommendations discussed in the methods section above. The `perm_se = FALSE` option is available primarily to match the behavior of other R packages that provide permutation tests that do not studentize by default (e.g., the permutation implementation in the "coin" R package).

```
set.seed(8812)
perm_t_test(
  formula = extra ~ group,
  data = sleep,
  paired = TRUE,
  # equivalence hypothesis
  alternative = "equivalence",
  # equivalence bounds
```

```
  mu = c(-0.5, 0.5),
  # number of random permutations
  R = 999
)
```

```
##
##  Randomization Permutation Paired t-test
##
## data:  extra by group
## t-observed = -2.7766, df = 9, p-value = 0.999
## alternative hypothesis: equivalence
## null values:
## difference in means difference in means
##                -0.5                 0.5
## 90 percent confidence interval:
##  -2.2607475 -0.8930238
## sample estimates:
## mean of the differences (z = '1' - '2')
##                                   -1.58
```

Running the bootstrap and permutation equivalence tests on the same paired sleep data using matching equivalence bounds produces broadly similar conclusions, as would be expected. The two approaches can, however, diverge in small samples, under strong heteroscedasticity, or in the presence of influential outliers, which is where the choice between them becomes substantively important. Users interested in robustness to outliers can additionally supply the `tr` argument to either function to use trimmed means, which downweight extreme observations while preserving a location-based interpretation on the original scale.

6.2.2.1. Checking Assumptions. The permutation t-test carries different assumptions depending on design. For the two-sample case, the studentized version is valid under exchangeability of observations across groups under the null and provides asymptotic validity even when variances differ between groups. For paired and one-sample designs, the sign-flip procedure additionally requires that the distribution of differences (or one-sample observations) is symmetric around the null value. Asymmetry makes the paired permutation t-test approximate rather than exact, but studentization provides some asymptotic protection. Symmetry is a commonly overlooked assumption for paired permutation tests and is easily checked with a symmetry plot of the differences, as demonstrated for the WMW signed-rank case above. For two-sample designs, overlaid empirical cumulative density functions and separate density plots help assess whether group shapes are comparable enough for exchangeability under the null to be credible. Diagnostic workflows for all three designs are demonstrated in the online supplement. As with the bootstrap, these checks reflect assumptions that are relaxed relative to Student's t-test but not eliminated.

6.3. *Log-Transformed Methods*

The natural log transformation stabilizes variance and approximates normality for right-skewed data (Bland and Altman 1996). An additional advantage is that differences on the log scale back-transform to ratios, making interpretation straightforward: if two samples have geometric means of 7 and 10.5, the ratio is 1.5 (i.e., 10.5/7). This property is central to pharmaceutical bioequivalence, where the FDA requires comparisons on the log-transformed scale with equivalence bounds of 0.80 to 1.25 for the ratio of geometric means (Food and Drug Administration 2014; He et al. 2022).

An alternative to `log_TOST` is to log-transform data manually and analyze with `t_TOST`, `simple_htest`, or the resampling methods. Differences on the log scale can then be interpreted as "sympercents" — symmetric percentage differences on the $100 \cdot \ln(x) - ln(y)$ scale (Cole 2000) — which are symmetric in a way that ordinary percentage differences are not because percentage changes are calculated from different bases. For example, a 10% loss and a 10% gain do not cancel out, whereas +10 and -10 sympercents are exactly equal in magnitude.[5]

6.3.1. Log TOST

The `log_TOST` function accepts paired and two-sample comparisons (one-sample tests are not supported since there is no ratio to calculate). Instead of an SMD, a ratio of means is reported (Lajeunesse 2015). The default equivalence bounds are set to the FDA standard (`eqb = 1.25`; the reciprocal is taken as the other bound).

```
log_TOST(mpg ~ am, data = mtcars)
```

```
##
## Log-transformed Welch Two Sample t-test
##
## The equivalence test was non-significant, t(23.96) = -1.363, p = 9.07e-01
## The null hypothesis test was significant, t(23.96) = -3.826, p = 8.19e-04
## NHST: reject null significance hypothesis that the effect is equal to one
## TOST: don't reject null equivalence hypothesis
##
## TOST Results
##                 t    df p.value
## t-test     -3.826 23.96 < 0.001
## TOST Lower -1.363 23.96   0.907
## TOST Upper -6.288 23.96 < 0.001
##
## Effect Sizes
##                   Estimate      SE               C.I. Conf. Level
## log(Means Ratio)  -0.3466 0.09061 [-0.5017, -0.1916]         0.9
## Means Ratio        0.7071      NA   [0.6055, 0.8256]         0.9
```

[5] A concrete illustration that I give my students: if a stock falls 10% on Tuesday and rises 10% from Tuesday on Wednesday, you still have a loss on of your original investment. Fortunately, sympercents do not have this asymmetry.

6.3.2. Bootstrap Log TOST

The bootstrap version uses the same bootstrap methods as `boot_t_test` on the log-transformed values:

```
boot_log_TOST(mpg ~ am, data = mtcars, R=999)
```

```
##
## Bootstrapped Log Welch Two Sample t-test
##
## The equivalence test was non-significant, t(23.96) = -1.363, p = 9.54e-01
## The null hypothesis test was significant, t(23.96) = -3.826, p = 0e+00
## NHST: reject null significance hypothesis that the effect is equal to 1
## TOST: don't reject null equivalence hypothesis
##
## TOST Results
##                  t    df p.value
## t-test     -3.826 23.96 < 0.001
## TOST Lower -1.363 23.96   0.954
## TOST Upper -6.288 23.96 < 0.001
##
## Effect Sizes
##                   Estimate      SE               C.I. Conf. Level
## log(Means Ratio)  -0.3466 0.08621 [-0.4932, -0.1893]         0.9
## Means Ratio        0.7071 0.06116   [0.6106, 0.8275]         0.9
## Note: studentized bootstrap ci method utilized.
## Equivalence Bounds: log(Means Ratio) [-0.2231, 0.2231]; Means Ratio [0.8, 1.25]
```

From this analysis, we would conclude there is a significant effect that is not practically equivalent.

6.3.3. Checking Assumptions

The log-transformed methods inherit the assumptions of the underlying test applied to the transformed data. The `log_TOST` function carries forward the t-test assumptions, and `boot_log_TOST` carries forward the bootstrap t-test assumptions. However, all these assumptions (i.e., normality, symmetry, and homogeneity of variance) should be assessed on the log scale rather than the original scale, since that is where inference now takes place. Similarly, the visual diagnostics should be computed on log-transformed data. This points to a broader practical use of log transformation. When the symmetry assumption of the sign-flip permutation test is violated because paired differences are right-skewed on the original scale, the differences of logs are often approximately symmetric, and `perm_t_test` on log-transformed values becomes a reasonable option. In such cases the equivalence bounds could be specified on the log scale and interpreted as log ratios (or sympercents) using any of the other t-test functions in TOSTER.

7. Equivalence Testing with ANOVAs

Many researchers use an ANOVA as an omnibus test before inspecting pairwise comparisons, particularly in factorial designs. As Campbell and Lakens (2021) note, the lack of a significant ANOVA result does not indicate that a factor in a design has no effect. Equivalence testing for ANOVAs addresses this directly by asking whether the variance attributable to a factor is small enough to be declared negligible.

7.1. *Partial Eta-Squared and the Equivalence Hypothesis*

The effect size used throughout this section is "partial eta-squared" (η_p^2),

$$\eta_p^2 = \frac{SS_{\text{effect}}}{SS_{\text{effect}} + SS_{\text{error}}}$$

The proportion of variance attributable to an effect once the variance attributable to the other effects in the model has been removed. It is not the same quantity as η^2, which divides by the total sum of squares and the two coincide only in a one-way design. In within-subjects designs, SS_{error} refers to the error term against which that particular effect is tested, so each effect in a factorial model carries its own denominator.

Because η_p^2 cannot be negative, there is no lower bound to test against. Oddly enough, this means the equivalence tests for ANOVAs and F-tests in TOSTER do not involve TOST. Instead it is only a single one-sided test, which Campbell and Lakens (2021) describe accurately as an "omnibus non-inferiority test".

$$H_0 : \eta_p^2 \geq \Delta \qquad \text{versus} \qquad H_1 : \eta_p^2 < \Delta$$

In this case, rejecting H_0 supports the claim that the effect in question explains less than the prespecified Δ of the variance. There are two caveats on that interpretation worth discussing. First, this equivalence test is a statement about an effect as a whole rather than about any particular contrast or comparison within it. An omnibus effect can be negligible while a specific pairwise comparison nested inside it is not, so equivalence at the level of the F-test does not license equivalence claims about individual group differences. Second, as with any equivalence test, failing to reject H_0 is not evidence that the effect is sufficiently large. It means only that the data remain compatible with an effect at or beyond the bound.

The test itself is computed from the cumulative distribution function of the non-central F distribution. Campbell and Lakens (2021) derived the p-value for a one-way ANOVA with J groups and N total observations as

$$p = p_F\left(F; J-1, N-J, \frac{N \cdot \Delta}{1-\Delta}\right)$$

and evaluated the Type I error and power of the procedure by simulation. With TOSTER I then generalized the non-centrality parameter so that the same logic extends to multi-factor, factorial, and within-subjects designs,

$$\lambda_{eq} = \frac{\Delta}{1-\Delta} \cdot (df_1 + df_2 + 1), \qquad p_{eq} = p_F(F; df_1, df_2, \lambda_{eq})$$

where df_1 and df_2 are the numerator and denominator degrees of freedom for the effect being tested. For a one-way design $df_1 + df_2 + 1 = (J-1) + (N-J) + 1 = N$, so the generalization reduces exactly to the expression given by Campbell and Lakens (2021). A fuller description is provided in the package documentation.

The quantity $df_1 + df_2 + 1$ is the effective sample size of the error stratum in which an effect is tested, which is not in general the total number of observations. In a mixed design, for example, it recovers the number of subjects for a between-subjects effect. I evaluated the generalization by simulation for one-way within-subjects, factorial between-subjects, and mixed designs, and the equivalence test maintained its nominal Type I error rate at the boundary of the null hypothesis in each. Readers who want the simulation code, the full results, and a longer discussion of interpretation will find them in the online supplement.

The equivalence test inherits the standard ANOVA assumptions, which should be checked on the fitted model itself rather than on the equivalence output. The `performance::check_model()` function provides a comprehensive visual diagnostic suite for most linear model objects in R in a single call; the online supplement demonstrates its use.

However, TOSTER's current implementation is sensitive to violations to the assumption of sphericity. For within-subjects factors with more than two levels, `equ_anova` currently uses the uncorrected univariate degrees of freedom and does not apply Greenhouse-Geisser or Huynh-Feldt corrections, even when the fitted object was created with a correction requested. My simulations show that the equivalence test can become substantially anti-conservative when sphericity is badly violated. Sphericity should therefore be assessed on the fitted model, and results treated cautiously when it is in doubt; the supplement quantifies the size of the problem.

7.2. *Choosing a Bound on* η_p^2

In my opinion, the bounds on η_p^2 are harder to justify than bounds on the original measurement scale (much like the SMD), and the difficulty is worth confronting before running an equivalence ANOVA test.

The first problem is that η_p^2 is not comparable across designs. Its denominator depends on which other effects appear in the model and, in within-subjects designs, on the error term against which the effect is tested. A η_p^2 of 0.06 for a one-way between-subjects factor and a η_p^2 of 0.06 for an interaction in a repeated-measures model are not estimates of the same quantity. A bound therefore cannot simply be borrowed from a published study unless that study used the same design and the same model. This may be more of a problem for repeated-measures designs where subject variance is excluded from the denominator entirely. A bound on η_p^2 is consequently not a bound on the share of total variance in the data, and the gap between the two widens as the number of levels of the within-subjects factor falls. The supplement quantifies the discrepancy.

The second problem is that a proportion of variance has little direct substantive meaning. Unlike a mean difference expressed in the units of the outcome, there is rarely an anchor-based or cost-benefit argument that delivers a number on this scale

directly. Researchers who find Cohen's f more intuitive can move between the two with $f^2 = \eta_p^2/(1-\eta_p^2)$, and the `power_eq_f` function can be used to determine the degrees of freedom required to achieve a given power for a specified bound, or the power attainable for a given design. The choosing-bounds section of the package's F-test vignette discusses the problem in more detail. Additionally, I would recommend that, in most situations where an ANOVA is utilized for multi-group or even factorial designs, that equivalence testing is often best accomplished with specific contrasts (example below).

The bound of $\Delta = 0.35$ used in both examples below is deliberately large. It was chosen so that the demonstrations produce interpretable output rather than because it represents a defensible threshold, and it should not be read as a default.

7.3. *Example with Summary Statistics*

Using the `InsectSprays` data and the `equ_ftest` function, which accepts summary F-test statistics directly:

```
data("InsectSprays")
aovtest = aov(count ~ spray, data = InsectSprays)
anova(aovtest)
```

```
## Analysis of Variance Table
##
## Response: count
##           Df Sum Sq Mean Sq F value    Pr(>F)
## spray      5 2668.8  533.77  34.702 < 2.2e-16 ***
## Residuals 66 1015.2   15.38
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
equ_ftest(
  # F statistic and degrees of freedom from the ANOVA table
  Fstat = 34.70228,
  df1 = 5,
  df2 = 66,
  # equivalence bound on partial eta-squared
  eqb = 0.35
)
```

```
##
##  Equivalence Test from F-test
##
## data:  Summary Statistics
## F = 34.702, df1 = 5, df2 = 66, p-value = 1
## 95 percent confidence interval:
##  0.5806263 0.7804439
## sample estimates:
## [1] 0.724439
```

Based on these results, there is a significant effect of "spray" and the differences are

not statistically equivalent. We then reject the traditional null hypothesis but fail to reject the equivalence null hypothesis.

7.4. *Example with R ANOVA Objects*

The `equ_anova` function accepts objects from `stats::aov`, `car::Anova`, and `afex::aov_car`, avoiding the need to extract summary statistics manually. It returns the familiar ANOVA table with the equivalence test appended:

- `effect`, `df1`, `df2`, `F.value` — the effect being tested and the components of the traditional F-test.
- `p.null` — the p-value for the traditional nil-hypothesis test.
- `pes` — the estimated η_p^2 for that effect.
- `eqbound` — the equivalence bound that was applied.
- `p.equ` — the p-value for the equivalence test.

Because each effect in a within-subjects design is tested against its own error term, each row carries its own df_2 and therefore its own denominator for `pes`. Note also that an equivalence test is reported for the intercept; this is rarely a hypothesis of interest and can generally be ignored.

```
# Example using a purely within-subjects design
# (Maxwell & Delaney, 2004, Chapter 12, Table 12.5, p. 578):
library(afex)
data(md_12.1)
aovtest2 = aov_ez(
  # participant identifier
  "id",
  # dependent variable
  "rt",
  # data
  md_12.1,
  # within-subjects factors
  within = c("angle", "noise"),
  anova_table = list(correction = "none", es = "none")
)
equ_anova(aovtest2, eqb = 0.35)
```

```
##        effect df1 df2   F.value       p.null       pes eqbound     p.equ
## 1 (Intercept)   1   9 598.44917 1.526600e-09 0.9851839    0.35 0.9999997
## 2       angle   2  18  40.71910 2.086763e-07 0.8189831    0.35 0.9992557
## 3       noise   1   9  33.76596 2.559737e-04 0.7895522    0.35 0.9763228
## 4 angle:noise   2  18  45.31034 9.424093e-08 0.8342857    0.35 0.9996103
```

7.5. *Example with Specific Contrasts*

As mentioned previously, the ANOVA equivalence testing functions answer only a very broad, somewhat vague, question. The omnibus test asks whether a factor as a whole explains a negligible share of the variance available to it. That is often not the question a researcher actually has. More commonly the hypothesis concerns particular comparisons (e.g., is this condition equivalent to that one?) and, as noted above, an omnibus effect can be negligible while a specific comparison nested within it is not. Testing contrasts directly also sidesteps the bound-justification problem described earlier. A contrast is expressed in the units of the outcome, so the smallest effect size of interest can be argued for on the original measurement scale, where anchor-based and cost-benefit reasoning are available, rather than on the proportion-of-variance scale where they usually are not.

TOSTER does not fit contrasts itself, but the "emmeans" package (Lenth and Piaskowski 2026) accommodates non-zero null hypotheses directly and operates on the same fitted model objects[6]. Therefore, I wanted to briefly demonstrate how this could be accomplished. Using the the `InsectSprays` ANOVA fitted above, suppose a difference of fewer than five insects is considered too small to matter and we have two specific hypotheses of equivalence regarding groups C, D, and E:

```
library(emmeans)
# estimated marginal means from the ANOVA fit earlier
emm = emmeans(aovtest, ~ spray)

# two comparisons specified in advance as the hypotheses of interest
sprays = contrast(
  emm,
  method = list(
    "C vs D" = c(0, 0, 1, -1, 0, 0),
    "C vs E" = c(0, 0, 1, 0, -1, 0)
  )
)

test(
  sprays,
  # equivalence bound on the original scale (insect counts)
  delta = 5,
  side = "equivalence",
  adjust = "none"
)
```

```
##  contrast estimate  SE df t.ratio p.value
##  C vs D      -2.83 1.6 66  -1.353  0.0903
##  C vs E      -1.42 1.6 66  -2.238  0.0143
##
## Statistics are tests of equivalence with a threshold of 5
## P values are left-tailed
```

Spray C is statistically equivalent to spray E ($p = 0.014$) but not to spray D (p

[6]This can also be accomplished with "marginaleffects" (Arel-Bundock, Greifer, and Heiss 2024)

= 0.090), even though the omnibus test treated "spray" as a single undifferentiated effect. The same conclusion can be read from the interval estimates, which for TOST are taken at $1 - 2\alpha$:

```
confint(sprays, level = 0.90)
```

```
##  contrast estimate  SE df lower.CL upper.CL
##  C vs D      -2.83 1.6 66    -5.50   -0.162
##  C vs E      -1.42 1.6 66    -4.09    1.254
##
## Confidence level used: 0.9
```

The 90% interval for C versus E lies entirely inside $\pm$ 5, whereas the interval for C versus D extends past the lower bound.

Two notes on this additional non-TOSTER derived workflow. First, whether to adjust for multiplicity is a substantive decision rather than an automatic one; no adjustment is applied above because the two contrasts were specified in advance as the hypotheses of interest (in this case we could consider these independent tests). However, multiplicity corrections (e.g., Holm-Bonferroni) remain valid for equivalence tests, and are worthy of consideration. Second, this approach reaches well beyond the ANOVA. Because `emmeans` works from fitted model objects, the same equivalence contrasts can be applied to mixed-effects models and other designs that fall outside TOSTER's scope.

8. Power Analysis and Sample Size Planning

Researchers planning an equivalence study will need to determine the required sample size. Power for equivalence tests behaves differently than for standard tests: larger samples are generally needed, and power depends on both the true effect size and the width of the equivalence bounds.

8.1. *Power for TOST t-tests*

The `power_t_TOST` function computes exact power for TOST with one-sample, two-sample, and paired t-tests. It can solve for any one of `n`, `power`, or `alpha` given the other parameters are provided. The `delta` argument specifies the true mean difference (set to 0 if the true effect is exactly zero) and `eqb` specifies the equivalence bounds.

For example, to determine the sample size needed to achieve 80% power to detect equivalence within bounds of $\pm$ 0.5 (assuming the true difference is zero and $SD = 1$):

```
power_t_TOST(
  # true mean difference
  delta = 0,
  # assumed standard deviation
  sd = 1,
  # equivalence bounds
  eqb = 0.5,
  alpha = 0.05,
  # leaving n unspecified solves for the n giving 80% power
  power = 0.8,
  type = "two.sample"
)
```

```
##
##      Two-sample TOST power calculation
##
##           power = 0.8
##            beta = 0.2
##           alpha = 0.05
##               n = 69.19782
##           delta = 0
##              sd = 1
##          bounds = -0.5, 0.5
##
## NOTE: n is number in *each* group
```

8.2. *Power for Correlations*

The `power_z_cor` function provides power analysis for equivalence tests on Pearson correlations using Fisher's z-transformation:

```r
power_z_cor(
  # true correlation
  rho = 0,
  # leaving n unspecified solves for the n giving 80% power
  power = 0.8,
  # equivalence bounds of -0.3 and 0.3
  null = 0.3,
  alpha = 0.05,
  # equivalence hypothesis
  alternative = "equivalence"
)
```

```
##
##      Approximate Power for Pearson Product-Moment Correlation (z-test)
##
##               n = 92.38991
##             rho = 0
##           alpha = 0.05
##            beta = 0.2
##           power = 0.8
##            null = 0.3, -0.3
##     alternative = equivalence
```

8.3. *Extensions: PowerTOST and Superpower*

For more complex designs (e.g., crossover studies, corssover-replicate designs, group-sequential equivalence trials) the PowerTOST package (Labes, Schütz, and Lang 2021) provides extensive functionality beyond what TOSTER offers. For factorial designs where equivalence contrasts are tested via estimated marginal means, simulation-based power analysis via the Superpower package is recommended (Lakens and Caldwell 2021). Both packages complement TOSTER's built-in power tools for common research designs. In some situations, customized simulations may be required to accurately determine power for an equivalence test.

9. Conclusions

This paper has described the major updates to the TOSTER R package, expanding its scope from the original t-test and correlation equivalence tests to a comprehensive toolkit for frequentist equivalence testing. The update adds a unified t-test interface with minimal effect testing support, a simplified hypothesis testing interface that integrates with the standard R ecosystem, tools for standardized effect sizes and correlations, robust and resampling-based alternatives, ANOVA equivalence testing, and power analysis for sample size planning.

Several limitations should be acknowledged. TOSTER only covers a few standard research designs, and it does not handle many analysis scenarios such as those involving multilevel or mixed-effects models. For these settings, researchers should consider packages like "emmeans" (Lenth and Piaskowski 2026) or "marginaleffects" (Arel-Bundock, Greifer, and Heiss 2024), which can accommodate non-zero null hypotheses in more complex modeling frameworks. TOSTER also does not provide Bayesian equivalence testing; researchers interested in Bayesian methods for the null should consult other R packages such as "bayestestR" (Makowski, Ben-Shachar, and Lüdecke 2019).

I hope that the updated package continues to build upon the original impact of Lakens (2017) and makes equivalence testing more accessible to researchers across disciplines. At the very least, I would encourage users to compare results across methods (e.g., Student's t-test vs. permutation t-test vs. bootstrap t-test) to ensure that conclusions do not entirely depend on the chosen analysis. If the interpretation substantially changes depending on the test, this suggests features of the data — violations of assumptions, outliers, distributional shape — that warrant further investigation.

10. Additional Information

The TOSTER package can be installed from CRAN or from GitHub:

```
# Install from CRAN
install.packages("TOSTER")

# Install the development version from GitHub
devtools::install_github("Lakens/TOSTER")
```

The supplementary materials for this article are archived on Zenodo (https://doi.org/10.5281/zenodo.22776188). They include a verification of TOSTER against the results reported by Campbell and Lakens (2021) and a set of assumption checks for the procedures described in this article. Each supplement is provided as a Quarto source file with its rendered output, and all analyses were conducted with TOSTER v0.9.0.

Acknowledgement(s)

I would like to thank everyone from the Lakens laboratory group for their input and suggestions. I must also thank David Eisner, Jakub Tomek, and James Selig for their feedback on drafts of this manuscript. Daniel Lakens provided a review of many of the materials that have been incorporated into the update of TOSTER, and was the original author of this package. Without his help and encouragement, the TOSTER package and this update would not exist.

AI Use Disclosure

During the preparation of the TOSTER package updates and this manuscript, the author used Claude Opus 4.5-5.0 (Anthropic) to assist with editing R code and drafting unit tests. For the manuscript text, Claude Opus was used only for targeted copy-editing — suggesting revisions to specific passages, catching typographical and grammatical errors, and identifying inconsistencies across sections. The manuscript was written by the author, and all AI-assisted output was reviewed, revised, and verified by the author, who takes full responsibility for the content of the package and the manuscript.

Nomenclature/Notation

- ANOVA: Analysis of Variance
- Bootstrapping: the use of random sampling with replacement to estimate statistics
- FDA: Food and Drug Administration (United States of America)
- MET: Minimal Effects Test
- ncp: non-centrality parameter
- SESOI: Smallest Effect Size of Interest
- SMD: Standardized Mean Difference (e.g., Cohen's d)
- TOST: Two One-Sided Tests
- WMW: Wilcoxon-Mann-Whitney

Notes

The R package is also (partially) implemented in jamovi as the TOSTER module.

Declarations

Funding

No funding was received for this work.

Conflicts of interest/Competing interests

The author is the maintainer of the TOSTER R package described in this manuscript. The package is free and open-source software, and the author receives no financial compensation related to it. The author declares no other competing interests.

Ethics approval

Not applicable. This article describes statistical software and uses only publicly available, de-identified example datasets distributed with R packages; no new data were collected from human participants or animals.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

No new data were generated for this article. All example datasets are publicly available within R packages: `bugs` (jmv package), `sleep` and `InsectSprays` (base R datasets package), and `md_12.1` (afex package). The supplementary materials, including the verification against Campbell and Lakens (2021) and the assumption-check workflows, are archived on Zenodo (https://doi.org/10.5281/zenodo.22776188). None of the analyses reported here were preregistered.

Code availability

The TOSTER R package (v0.9.0) is freely available under the GPL-3 license from CRAN (https://CRAN.R-project.org/package=TOSTER) and GitHub (https://github.com/Lakens/TOSTER). The R code needed to reproduce all examples is included in the manuscript, and the Quarto source files for the supplementary materials are archived on Zenodo (https://doi.org/10.5281/zenodo.22776188).

Authors' contributions

A.R.C. is the sole author and was responsible for conceptualization, software development, formal analysis, visualization, and writing (original draft and review & editing).

References

Altman, Douglas G, and J Martin Bland. 1995. "Statistics notes: Absence of evidence is not evidence of absence." *BMJ* 311 (7003): 485. https://doi.org/10.1136/bmj.311.7003.485.

Anvari, Farid, and Daniël Lakens. 2021. "Using anchor-based methods to determine the smallest effect size of interest." *Journal of Experimental Social Psychology* 96: 104159.

Arboretti, Rosa, Fortunato Pesarin, and Luigi Salmaso. 2020. "A unified approach to permutation testing for equivalence." *Statistical Methods & Applications* 30 (3): 1033–1052. http://dx.doi.org/10.1007/s10260-020-00548-0.

Arel-Bundock, Vincent, Noah Greifer, and Andrew Heiss. 2024. "How to Interpret Statistical Models Using marginaleffects for R and Python." *Journal of Statistical Software* 111 (9): 1–32.

Baguley, Thom. 2009. "Standardized or simple effect size: What should be reported?" *British Journal of Psychology* 100 (3): 603–617.

Bland, J Martin, and Douglas G Altman. 1996. "Statistics notes: the use of transformation when comparing two means." *BMJ* 312 (7039): 1153. https://doi.org/10.1136/bmj.312.7039.1153.

Brunner, Edgar, and Ullrich Munzel. 2000. "The nonparametric Behrens-Fisher problem: Asymptotic theory and a small-sample approximation." *Biometrical Journal* 42 (1): 17–25. https://doi.org/10.1002/(SICI)1521-4036(200001)42:1<17::AID-BIMJ17>3.0.CO;2-U.

Bürkner, Paul-Christian, and Matti Vuorre. 2019. "Ordinal Regression Models in Psychology: A Tutorial." *Advances in Methods and Practices in Psychological Science* 2 (1): 77–101. http://dx.doi.org/10.1177/2515245918823199.

Caldwell, Aaron, and Andrew D. Vigotsky. 2020. "A case against default effect sizes in sport and exercise science." *PeerJ* 8: e10314. https://doi.org/10.7717/peerj.10314.

Campbell, Harlan, and Daniël Lakens. 2021. "Can we disregard the whole model? Omnibus non-inferiority testing for R2 in multi-variable linear regression and in ANOVA." *British Journal of Mathematical and Statistical Psychology* 74 (1): e12201. https://bpspsychub.onlinelibrary.wiley.com/doi/abs/10.1111/bmsp.12201.

Chung, EunYi, and Joseph P Romano. 2013. "Exact and asymptotically robust permutation tests." *The Annals of Statistics* 41 (2): 484–507. https://doi.org/10.1214/13-AOS1090.

Cole, TJ. 2000. "Sympercents: Symmetric percentage differences on the 100 log_e scale simplify the presentation of log transformed data." *Statistics in Medicine* 19 (22): 3109–3125. https://doi.org/10.1002/1097-0258(20001130)19:22<3109::AID-SIM558>3.0.CO;2-F.

Cook, Jonathan A, Steven A Julious, William Sones, Lisa V Hampson, Catherine Hewitt, Jesse A Berlin, Deborah Ashby, et al. 2018. "DELTA2 guidance on choosing the target difference and undertaking and reporting the sample size calculation for a randomised controlled trial." *BMJ* k3750. http://dx.doi.org/10.1136/bmj.k3750.

Divine, George W, H James Norton, Anna E Barón, and Elizabeth Juarez-Colunga. 2018. "The Wilcoxon–Mann–Whitney procedure fails as a test of medians." *The American Statistician* 72 (3): 278–286. https://doi.org/10.1080/00031305.2017.1305291.

Efron, Bradley, and Robert J. Tibshirani. 1993. *An Introduction to the Bootstrap*. Monographs on Statistics and Applied Probability 57. Boca Raton, Florida, USA: Chapman & Hall/CRC.

Fay, Michael P., and Yaakov Malinovsky. 2018. "Confidence intervals of the Mann-Whitney parameter that are compatible with the Wilcoxon-Mann-Whitney test." *Statistics in Medicine* 37 (27): 3991–4006. http://dx.doi.org/10.1002/sim.7890.

Food and Drug Administration. 2014. *Bioavailability and Bioequivalence Studies Submitted in NDAs or INDs — General Considerations*. Vol. FDA-2014-D-0204. Center for Drug Evaluation and Research. https://www.fda.gov/regulatory-information/search-fda-guidance-documents/bioavailability-and-bioequivalence-studies-submitted-ndas-or-in.

Funder, David C., and Daniel J. Ozer. 2019. "Evaluating Effect Size in Psychological Research:

Sense and Nonsense." *Advances in Methods and Practices in Psychological Science* 2 (2): 156–168.

Greenland, Sander, Malcolm Maclure, James J. Schlesselman, Charles Poole, and Hal Morgenstern. 1991. "Standardized regression coefficients: A further critique and review of some alternatives." *Epidemiology* 2 (5): 387–392.

He, Y, Y Deng, C You, and X H Zhou. 2022. "Equivalence tests for ratio of means in bioequivalence studies under crossover designs." *Statistical Methods in Medical Research* 09622802221093721. https://doi.org/10.1177/09622802221093721.

Jacobson, Neil S., and Paula Truax. 1991. "Clinical significance: A statistical approach to defining meaningful change in psychotherapy research." *Journal of Consulting and Clinical Psychology* 59 (1): 12–19. http://dx.doi.org/10.1037/0022-006X.59.1.12.

Janssen, Arnold. 1997. "Studentized permutation tests for non-i.i.d. hypotheses and the generalized Behrens-Fisher problem." *Statistics & Probability Letters* 36 (1): 9–21. https://doi.org/10.1016/S0167-7152(97)00043-6.

Karch, Julian D. 2021. "Psychologists should use Brunner-Munzel's instead of Mann-Whitney's U test as the default nonparametric procedure." *Advances in Methods and Practices in Psychological Science* 4 (2): 2515245921999602. https://doi.org/10.1177/2515245921999602.

Kirby, Kris N., and Daniel Gerlanc. 2013. "BootES: An R package for bootstrap confidence intervals on effect sizes." *Behavior Research Methods* 45 (4): 905–927. https://doi.org/10.3758/s13428-013-0330-5.

Labes, Detlew, Helmut Schütz, and Benjamin Lang. 2021. *PowerTOST: Power and Sample Size for (Bio)Equivalence Studies*. R package version 1.5-3, https://CRAN.R-project.org/package=PowerTOST.

Lajeunesse, Marc J. 2015. "Bias and correction for the log response ratio in ecological meta-analysis." *Ecology* 96 (8): 2056–2063. https://doi.org/10.1890/14-2402.1.

Lakens, Daniël. 2017. "Equivalence tests: A practical primer for t-tests, correlations, and meta-analyses." *Social Psychological and Personality Science* 1: 1–8. https://doi.org/10.1177/1948550617697177.

Lakens, Daniël, Neil McLatchie, Peder M Isager, Anne M Scheel, and Zoltan Dienes. 2020. "Improving inferences about null effects with Bayes factors and equivalence tests." *The Journals of Gerontology: Series B* 75 (1): 45–57. https://doi.org/10.1093/geronb/gby065.

Lakens, Daniël, Anne M Scheel, and Peder M Isager. 2018. "Equivalence testing for psychological research: A tutorial." *Advances in Methods and Practices in Psychological Science* 1 (2): 259–269. https://doi.org/10.1177/2515245918770963.

Lakens, Daniël, and Aaron R. Caldwell. 2021. "Simulation-Based Power Analysis for Factorial Analysis of Variance Designs." *Advances in Methods and Practices in Psychological Science* 4 (1). http://dx.doi.org/10.1177/2515245920951503.

Lenth, Russell V., and Julia Piaskowski. 2026. *emmeans: Estimated Marginal Means, aka Least-Squares Means*. R package version 2.0.2, https://CRAN.R-project.org/package=emmeans.

Liddell, Torrin M., and John K. Kruschke. 2018. "Analyzing ordinal data with metric models: What could possibly go wrong?" *Journal of Experimental Social Psychology* 79: 328–348. http://dx.doi.org/10.1016/j.jesp.2018.08.009.

Lundberg, Ian, Rebecca Johnson, and Brandon M. Stewart. 2021. "What Is Your Estimand? Defining the Target Quantity Connects Statistical Evidence to Theory." *American Sociological Review* 86 (3): 532–565. http://dx.doi.org/10.1177/00031224211004187.

Makowski, Dominique, Mattan S. Ben-Shachar, and Daniel Lüdecke. 2019. "bayestestR: Describing Effects and their Uncertainty, Existence and Significance within the Bayesian Framework." *Journal of Open Source Software* 4 (40): 1541. https://joss.theoj.org/papers/10.21105/joss.01541.

Mazzolari, Raffaele, Simone Porcelli, David J Bishop, and Daniël Lakens. 2022. "Myths and methodologies: The use of equivalence and non-inferiority tests for interventional studies in exercise physiology and sport science." *Experimental Physiology* 107 (3): 201–212. https://doi.org/10.1113/EP090171.

Neubert, Karin, and Edgar Brunner. 2007. "A studentized permutation test for the non-parametric Behrens-Fisher problem." *Computational Statistics & Data Analysis* 51 (10): 5192–5204. https://doi.org/10.1016/j.csda.2006.05.024.

Panzarella, Emily, Nataly Beribisky, and Robert A. Cribbie. 2021. "Denouncing the Use of Field-Specific Effect Size Distributions to Inform Magnitude." *PeerJ* 9: e11383.

Phipson, Belinda, and Gordon K Smyth. 2010. "Permutation p-values should never be zero: Calculating exact p-values when permutations are randomly drawn." *Statistical Applications in Genetics and Molecular Biology* 9 (1). https://doi.org/10.2202/1544-6115.1585.

Rafi, Zad, and Sander Greenland. 2020. "Semantic and cognitive tools to aid statistical science: replace confidence and significance by compatibility and surprise." *BMC medical research methodology* 20 (1): 1–13. https://doi.org/10.1186/s12874-020-01105-9.

Scheel, Anne M., Leonid Tiokhin, Peder M. Isager, and Daniël Lakens. 2020. "Why Hypothesis Testers Should Spend Less Time Testing Hypotheses." *Perspectives on Psychological Science* 16 (4): 744–755. http://dx.doi.org/10.1177/1745691620966795.

Schuirmann, Donald J. 1987. "A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability." *Journal of pharmacokinetics and biopharmaceutics* 15 (6): 657–680. https://doi.org/10.1007/BF01068419.

Thulin, Måns. 2024. "Modern Statistics with R." http://dx.doi.org/10.1201/9781003401339.

Tomek, Jakub, Aaron Caldwell, and David A. Eisner. 2026. "Beyond "non-significant" results: Why and how to test for practical equivalence." *Proceedings of the National Academy of Sciences* 123 (33). http://dx.doi.org/10.1073/pnas.2611548123.

Tukey, John W. 1969. "Analyzing data: Sanctification or detective work?" *American Psychologist* 24 (2): 83–91.

Viechtbauer, Wolfgang. 2007. "Approximate Confidence Intervals for Standardized Effect Sizes in the Two-Independent and Two-Dependent Samples Design." *Journal of Educational and Behavioral Statistics* 32 (1): 39–60. http://dx.doi.org/10.3102/1076998606298034.

Wilcox, Rand R. 1993. "Some results on a Winsorized correlation coefficient." *British Journal of Mathematical and Statistical Psychology* 46 (2): 339–349. http://dx.doi.org/10.1111/j.2044-8317.1993.tb01020.x.

Wilcox, Rand R. 1994. "The Percentage Bend Correlation Coefficient." *Psychometrika* 59 (4): 601–616. http://dx.doi.org/10.1007/BF02294395.